

Uncertainty

In the previous chapter we saw how variation arises in data generated by a model. We now confront a central issue: how to transform knowledge of this variation into statements about the uncertainty surrounding the model parameters. Uncertainty is a more elusive concept than variation and there is more disagreement about how it should be expressed. One important approach described in Chapter 11 uses Bayes' theorem to convert prior knowledge into posterior uncertainty, conditional on the data observed. The route taken below is more common in applications, and is usually known as the *frequentist, repeated sampling, or classical* approach. The next section describes how uncertainty may be expressed in terms of confidence intervals. In practice confidence intervals are usually approximate, but exact inferences are possible from some central models derived from the normal distribution, and these are described in the following section, followed by a brief summary of methods for prediction. There follows an introduction to the use of simulated data to appreciate both variation and uncertainty, for example in assessing the quality of approximate confidence intervals.

3.1 Confidence Intervals

3.1.1 Standard errors and pivots

In Section 2.2 we saw that many statistics approach limiting distributions in large samples. In practice a sample size is never infinite, but nevertheless these limits may be used to help quantify uncertainty. Suppose that T is an estimator of a parameter ψ based on a random sample $Y_1, \dots, Y_n$, that its unknown variance $\text{var}(T)$ has form τ^2/n , and that nV is a consistent estimator of τ^2 , so $nV \xrightarrow{P} \tau^2$ as $n \rightarrow \infty$. Statements of uncertainty about an estimator often involve its standard deviation $n^{-1/2}\tau$, but usually τ is unknown and must be estimated. An estimated standard deviation is known as a *standard error*, so $V^{1/2}$ is a standard error for T .

Example 3.1 (Average) Suppose each of the Y_j has mean μ and variance σ^2 . The sample average, $\bar{Y}$, has mean μ and variance σ^2/n . Now $S^2 = (n-1)^{-1} \sum (Y_j - \bar{Y})^2$ is an estimator of σ^2 , so $V^{1/2} = n^{-1/2}S$ is a standard error for $\bar{Y}$. ■

Example 3.2 (Gamma shape) In Example 2.26 we saw that the shape parameter κ of a gamma random sample may be estimated by $T = \bar{Y}^2/S^2$, and that this estimate has variance approximately $2\kappa(\kappa + 1)/n$, which may itself be estimated by $V = 2T(T + 1)/n$. We saw also that for the $n = 95$ observations in Example 2.3, $\bar{y}^2/s^2 = 3.15$. It follows that a standard error for this estimate is

$$\{2 \times 3.15(3.15 + 1)/95\}^{1/2} = 0.52.$$

■

In statements of uncertainty for an unknown parameter ψ , a central role is played by a *pivot* — a function of the data and the parameter whose distribution is known. In the discussion below we must distinguish a generic value of ψ from its true but unknown value ψ_0 . The condition that a quantity $Z(\psi_0)$ be pivotal means that for each z $\Pr\{Z(\psi_0) \leq z\}$ is the same for every ψ_0 ; that is, the distribution of $Z(\psi_0)$ does not depend on ψ_0 .

Example 3.3 (Exponential sample) Let $Y_1, \dots, Y_n$ be a random sample from the exponential distribution $1 - \exp(-y/\psi_0)$, $y > 0$, where $\psi_0 > 0$ is unknown. Then Y_j/ψ_0 has distribution function

$$\Pr(Y_j/\psi_0 \leq u) = \Pr(Y_j \leq u\psi_0) = 1 - \exp(-u),$$

which is known, even though the distribution of Y_j itself is not. Each of the Y_j/ψ_0 has this same distribution, and they are independent, so the distribution of $Z(\psi_0) = \psi_0^{-1} \sum Y_j$ is known, at least in principle. In fact the density of $Z(\psi_0)$ is $z^{n-1} \exp(-z)/(n-1)!$ for $z > 0$; this is the gamma density (2.7) with parameters $\lambda = 1$ and $\kappa = n$. As n is known, every property of the distribution of $Z(\psi_0)$ may be obtained. ■

Exact pivots are rare, but approximate ones are legion. For example, let $Z(\psi_0) = (T - \psi_0)/V^{1/2}$ be based on a sample of size n , and suppose that the limiting distribution of $Z(\psi_0)$ as $n \rightarrow \infty$ is standard normal; the results of Chapter 2 suggest that this will often be the case if T is based on averages. Then if n is large, $Z(\psi_0)$ is roughly standard normal, and so is an approximate pivot. Now

$$\Pr\{Z(\psi_0) \leq z\} = \Pr\left(\frac{T - \psi_0}{V^{1/2}} \leq z\right) \doteq \Phi(z),$$

where Φ is the standard normal distribution function. Then

$$\Pr\left(z_\alpha \leq \frac{T - \psi_0}{V^{1/2}} \leq z_{1-\alpha}\right) \doteq 1 - 2\alpha, \quad (3.1)$$

where z_α is the α quantile of this distribution, that is, $\Phi(z_\alpha) = \alpha$. Equivalently

$$\Pr(T - V^{1/2}z_{1-\alpha} \leq \psi_0 \leq T - V^{1/2}z_\alpha) \doteq 1 - 2\alpha. \quad (3.2)$$

Hence the random interval whose endpoints are

$$T - V^{1/2}z_{1-\alpha}, \quad T - V^{1/2}z_\alpha \quad (3.3)$$

contains ψ_0 with probability approximately $(1 - 2\alpha)$, whatever the value of ψ_0 . This interval is variously called an approximate $(1 - 2\alpha) \times 100\%$ confidence interval for

ψ_0 or a confidence interval for ψ_0 with approximate *coverage probability* $(1 - 2\alpha)$; we call it a $(1 - 2\alpha)$ confidence interval for ψ_0 . We regard the interval as random, containing ψ_0 with a specified probability. Conventionally α is a number such as 0.1, 0.05, 0.025, or 0.005, corresponding to 0.8, 0.9, 0.95 and 0.99 confidence intervals for ψ_0 ; these intervals will be increasingly wide. As $z_\alpha = -z_{1-\alpha}$, (3.3) may be written $T \pm V^{1/2} z_\alpha$. When $1 - 2\alpha = 0.95$, $z_\alpha = -1.96 \doteq -2$, so (3.3) is roughly $T \pm 2V^{1/2}$.

Given a particular set of data, $y_1, \dots, y_n$, we calculate the confidence interval from (3.3) by replacing T and V with their observed values t and v ; this gives $t \pm v^{1/2} z_\alpha$. This interval either does or does not contain ψ_0 , though we do not know which in any particular case. We interpret this by reference to a hypothetical infinite sequence of sets of data generated by the same mechanism or experiment that gave the data from which the interval was calculated. We then argue that if the observed data had been selected at random from these sets of data, then the interval actually obtained could be regarded as being selected randomly from a sequence of intervals with the property (3.2), and in this sense it would contain ψ_0 with probability $(1 - 2\alpha)$. With this interpretation, on average 19 out of every 20 confidence intervals with coverage 0.95 will contain ψ_0 , and on average 99 out of every 100 intervals with coverage 0.99 will contain ψ_0 , and so forth. Such an interval will also contain other values of ψ , but we would like it to be as short as possible on average, so that it does not contain too many of them.

Example 3.4 (Birth data) We use the data from Example 2.3 to construct a 95% confidence interval for the population mean time in the delivery suite, μ_0 hours, assuming that the times for each day are a random sample $Y_1, \dots, Y_n$ from the population.

An obvious choice of estimator T is the average, $\bar{Y}$, and we may take V to equal $n^{-1}S^2 = \{n(n-1)\}^{-1} \sum (Y_j - \bar{Y})^2$. In this case a $(1 - 2\alpha) \times 100\%$ confidence interval has endpoints $\bar{Y} \pm n^{-1/2} S z_\alpha$, and if $(1 - 2\alpha) = 0.95$, then $\alpha = 0.025$ and $z_\alpha = -1.96$. On day 1 there were $n = 16$ deliveries, with average $\bar{y} = 8.77$ and sample variance $s^2 = 18.46$, so a 95% confidence interval for μ_0 based on these data is $\bar{y} \pm n^{-1/2} s z_{0.025} = (6.66, 10.87)$ hours.

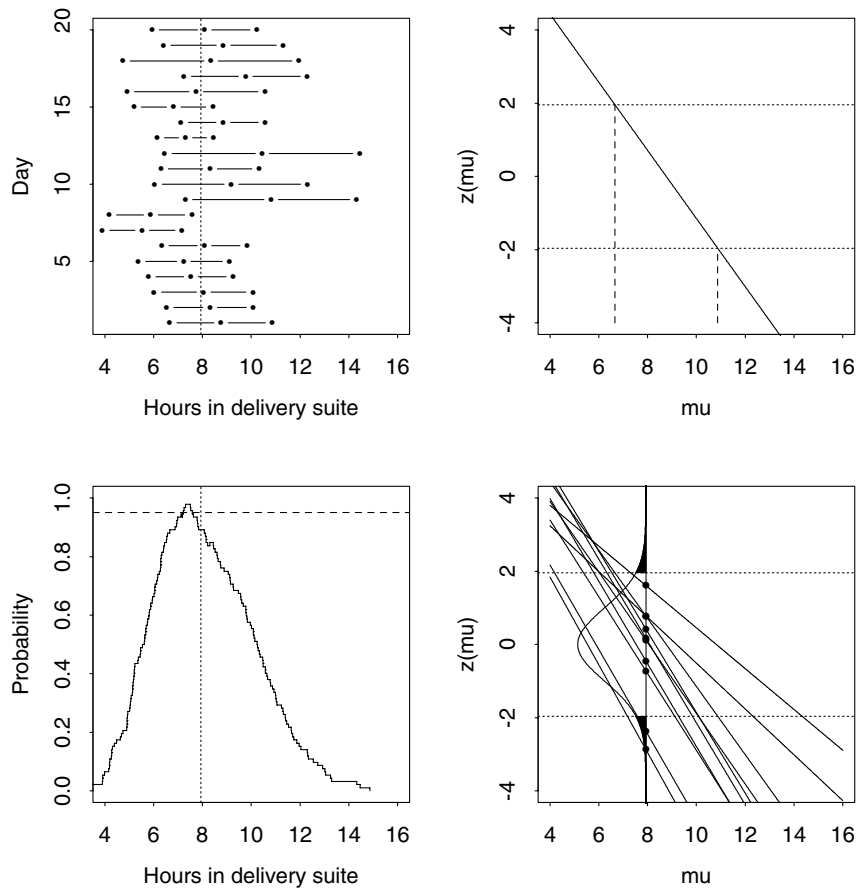
The upper left panel of Figure 3.1 shows 95% confidence intervals for μ_0 based on data for each of the first 20 days. The dotted line shows the average time in the delivery suite for all three months of data, which should be close to μ_0 . The intervals vary in length and in location, with 18 of them containing the three-month average. We expect about 19 of these 20 intervals to contain the true parameter, and the data seem consistent with this.

The upper right panel illustrates the calculation of the confidence interval from the day 1 data. The horizontal axis shows values of μ , and the diagonal line shows the function $z(\mu) = (8.77 - \mu)/(18.46/16)^{1/2}$. The confidence interval is obtained by reading off those values of μ for which $z(\mu) = z_{0.025}, z_{0.975} = \pm 1.96$, and these are shown by the vertical dashed lines, values of μ between which lie in the interval.

Other values of $\bar{Y}$ and S^2 that might have been observed would give different functions $Z(\mu) = (\bar{Y} - \mu)/(S^2/n)^{1/2}$. The lower right panel shows the observed values $z(\mu)$ of these for each of the first ten days of data. An infinite number of days would induce a probability density for $Z(\mu_0)$, corresponding to the points where the solid

Figure 3.1 Confidence intervals for the mean time in the delivery suite.

Upper left: 95% confidence intervals calculated using each of the first 20 days of data, with the average time for three months (92 days) of data (dots). Upper right: $z(\mu) = (\bar{y} - \mu)/(s^2/n)^{1/2}$ as a function of μ for the data from day 1 (diagonal line). The dotted lines show $z_{0.025} = -1.96$ and $z_{0.975} = 1.96$, from which the confidence interval is read off by solving $z(\mu) = \pm 1.96$. Lower right: lines $z(\mu)$ for ten different samples; their intersections $z(\mu_0)$ with the vertical line at μ_0 (blobs) have the standard normal density shown. If μ_0 were different, the density would be translated in the x -direction but remain unchanged, because $Z(\mu_0)$ is a pivot. Lower left: proportion of all 92 95% confidence intervals that include different values of μ . The vertical line (dots) shows the most likely value of μ_0 , where the coverage probability should be 0.95, given by the horizontal line (dashes).



vertical line intersects with the diagonal lines, and this density is illustrated also. If μ_0 was equal to the three-month average of 7.93 hours, we would expect a proportion 0.025 of the blobs at $z(7.93)$ to lie outside ± 1.96 . Exact pivotality of $Z(\mu_0)$ would mean that even if μ_0 was not 7.93 hours, so that the density was shifted horizontally, it would not change shape. In fact the normal approximation is not perfect here, as we shall see in Example 3.6.

We can compute the probability that the confidence interval (3.3) contains any value of μ . For μ_0 this should be $(1 - 2\alpha)$, but it will be lower for other values of μ . The lower left panel of Figure 3.1 shows the proportion of the 92 separate daily 95% confidence intervals containing each value of μ . This shows the shape we would expect: values close to the three-month average lie in most of the intervals, while values far from it are rarely covered. The corresponding proportions from an infinite number of days of data are the coverage probabilities

$$\Pr(T - z_{1-\alpha}V^{1/2} \leq \mu \leq T - z_{\alpha}V^{1/2} \mid \text{true value is } \mu_0).$$

If the approximation (3.2) was perfect, this probability would equal 0.95 when $\mu = \mu_0$, but a poor approximation would give a probability different from 0.95. We would

hope that this function would be as peaked as possible, to reduce the probability that a value other than μ_0 is contained in the interval: we want the average length of the intervals to be as short as possible. ■

Example 3.5 (Binomial distribution) In opinion polls about the status of the political parties in the UK, $m = 1000$ people are typically asked about their voting intentions. Let the number of these who support a particular party be denoted by R , supposed binomial with probability π . An estimate of π is $\hat{\pi} = R/m$, and since $\hat{\pi}$ has variance $\pi(1 - \pi)/m$, the standard error of $\hat{\pi}$ is $\{\hat{\pi}(1 - \hat{\pi})/m\}^{1/2}$. Example 2.17 combined with Slutsky's lemma (2.15) implies that $(\hat{\pi} - \pi)/\{\hat{\pi}(1 - \hat{\pi})/m\}^{1/2}$ converges in distribution to a standard normal variable, and consequently a $(1 - 2\alpha)$ confidence interval for π has endpoints

$$\hat{\pi} - z_{1-\alpha}\{\hat{\pi}(1 - \hat{\pi})/m\}^{1/2}, \quad \hat{\pi} - z_{\alpha}\{\hat{\pi}(1 - \hat{\pi})/m\}^{1/2}.$$

For the two main parties π usually lies in the range 0.3–0.4, so suppose that $\hat{\pi} = 0.35$, $m = 1000$, and we want a 95% confidence interval for π , so that $z_{0.975} = -z_{0.025} = 1.96 \doteq 2$. Then as $(0.35 \times 0.65/1000)^{1/2} \doteq 0.015$, the interval lies roughly 0.03 on either side of $\hat{\pi}$. In percentage terms this is the ‘3% margin of error’ sometimes mentioned when the results of such a poll are reported. The margin depends little on $\hat{\pi}$ because the function $\pi(1 - \pi)$ is fairly flat over the usual range 0.2–0.5 of support for the main parties. ■

There are infinitely many confidence intervals with coverage $(1 - 2\alpha)$, because we can replace $z_{1-\alpha}$ and z_{α} in (3.3) with any pair $z_{1-\alpha_1}$, z_{α_2} such that $\alpha_1, \alpha_2 \geq 0$ and $1 - \alpha_1 - \alpha_2 = 1 - 2\alpha$. The choice $\alpha_1 = \alpha_2 = \alpha$ gives the *equi-tailed* intervals discussed above, and these are common in practice. Other standard choices are $\alpha_1 = 2\alpha$, $\alpha_2 = 0$ or $\alpha_1 = 0$, $\alpha_2 = 2\alpha$, which give *one-sided* intervals $(T - V^{1/2}z_{1-2\alpha}, \infty)$ or $(-\infty, T - V^{1/2}z_{2\alpha})$ respectively. These are appropriate when a lower or an upper confidence bound is required for ψ_0 . For example, insurance companies are interested in upper confidence bounds for potential losses, lower bounds being of little interest.

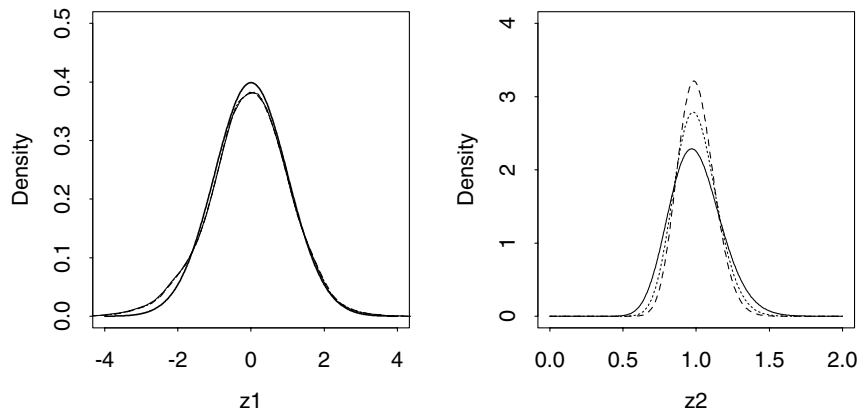
Complications

In order not to obscure the main points, the discussion above has been deliberately oversimplified. One complication is that realistic models rarely have just one parameter, so our notion of a pivot must be generalized.

Suppose that in addition to ψ , the model has another parameter λ whose value is not of interest, and that we seek to construct a confidence interval for ψ_0 using a pivot $Z(\psi_0)$. Our previous definition must be extended to mean that the distribution of $Z(\psi_0)$ depends neither on ψ_0 nor on λ . This is a stronger requirement than before and harder to satisfy.

A second complication is that there may be several possible (approximate) pivots, so that some basis is needed for choosing the best of them. Obviously we would like a pivot whose distribution depends as little as possible on the parameters, and preferably one that is exact, but we should also like short confidence intervals and a reliable general procedure for obtaining them. We describe some such procedures in

Figure 3.2 Densities of two approximate pivots for setting confidence intervals for the gamma mean, based on samples of size $n = 15$ from the gamma distribution. Left panel: density estimates based on 10,000 values of $Z_1(\mu_0) = n^{1/2}(\bar{Y} - \mu_0)/S$, for shape parameter $\kappa = 2$ (solid), 3 (dots), 4 (dashes), with $N(0, 1)$ density (heavy). Right panel: density of $Z_2(\mu_0) = \bar{Y}/\mu_0$ for $\kappa = 2$ (line), 3 (dots) 4 (dashes).



Chapter 4, and return to a general discussion in Chapter 7. The following example illustrates some of the difficulties.

Example 3.6 (Gamma distribution) A random variable Y with gamma density (2.8) may be expressed as $Y = \mu X$, where X has density (2.8) with $\mu = 1$, that is, it has unit mean and shape parameter κ . If $Y_1, \dots, Y_n$ is a sample from the gamma density with parameters μ_0 and κ , then

$$Z_1(\mu_0) = \frac{\bar{Y} - \mu_0}{\left\{ \frac{1}{n(n-1)} \sum (Y_j - \bar{Y})^2 \right\}^{1/2}} = \frac{\bar{X} - 1}{\left\{ \frac{1}{n(n-1)} \sum (X_j - \bar{X})^2 \right\}^{1/2}},$$

and hence the distribution of $Z_1(\mu_0)$ is independent of μ_0 . As $n \rightarrow \infty$, $Z_1(\mu_0) \xrightarrow{D} N(0, 1)$, giving the confidence interval (3.3), but for any given n the distribution of $Z_1(\mu_0)$ depends on n and on κ . Estimates of this density for $n = 16$ and $\kappa = 2, 3$, and 4 are shown in the left panel of Figure 3.2. The density seems stable over κ , but it is skewed to the left compared to the limiting normal density. Thus although $Z_1(\mu_0)$ appears to be roughly pivotal, values of the normal quantiles z_α might not give good confidence bounds; this would chiefly affect the upper limit.

Another possible pivot here is $Z_2(\mu_0) = \bar{Y}/\mu_0 = \bar{X}$, which turns out to have the gamma density (2.8) with unit mean and shape parameter $n\kappa$. Let $g_\alpha(n\kappa)$ be the α quantile of this distribution. Then

$$\begin{aligned} 1 - 2\alpha &= \Pr\{g_\alpha(n\kappa) \leq \bar{Y}/\mu_0 \leq g_{1-\alpha}(n\kappa)\} \\ &= \Pr\{\bar{Y}/g_{1-\alpha}(n\kappa) \leq \mu_0 \leq \bar{Y}/g_\alpha(n\kappa)\}, \end{aligned}$$

giving a $(1 - 2\alpha)$ confidence interval $(\bar{y}/g_{1-\alpha}(n\kappa), \bar{y}/g_\alpha(n\kappa))$ based on a sample $y_1, \dots, y_n$. In practice κ is unknown and must be replaced by an estimate $\hat{\kappa}$, so $Z_2(\mu_0)$ is also an approximate pivot.

Consider the day 1 data for the delivery suite, for which $n = 16$, $\bar{y} = 8.77$ and suppose $\hat{\kappa} = 3$. With $\alpha = 0.025$ we find that $g_\alpha(n\hat{\kappa}) = 0.737$, $g_{1-\alpha}(n\hat{\kappa}) = 1.302$. This gives 95% confidence interval (6.74, 11.89) hours for μ_0 . This interval is longer than that given by the pivot $Z_1(\mu_0)$, (6.66, 10.87), and it is not symmetric about $\bar{y}$.

Densities for $Z_2(\mu_0)$ shown in the right panel of Figure 3.2 depend much more on κ than those for $Z_1(\mu_0)$. Thus here we have a choice between two approximate pivots, one which is close to pivotal but whose distribution can only be estimated, and another which is further from pivotal but whose quantiles are known. ■

Interpretation

The repeated sampling basis for interpretation of confidence intervals is not universally accepted. The central issue is whether or not hypothetical repetitions bear any relevance to the data actually obtained. One view is that since every set of data is unique, such repetitions would be irrelevant even if they existed, and another basis must be found for statements of uncertainty; see Chapter 11. However it is reassuring that intervals derived from different principles are often similar and sometimes identical for standard problems, and in practice most users do not worry greatly about the precise interpretation of the uncertainty measures they report. The essential point is to provide some assessment of uncertainty, as honest as possible.

Another view is that the repeated sampling interpretation is secure provided the hypothetical data contain the same information, defined suitably, as the original data, but that if the set of hypothetical datasets taken is too large then it is irrelevant to the data actually observed. Thus in the delivery suite example we might argue that as day 1 had 16 arrivals, the relevant hypothetical repetitions are for days with 16 arrivals, because to know the number of arrivals is informative about the precision of any parameter estimate, though not about its value.

3.1.2 Choice of scale

The delta method provides standard errors and limiting distributions for smooth functions of random variables. This poses a problem, however: on what scale should a confidence interval for ψ_0 be calculated? For suppose that h is a monotone function, and that (L, U) is a $(1 - 2\alpha)$ confidence interval for $h(\psi_0)$, that is, $\Pr\{L \leq h(\psi_0) \leq U\} \doteq 1 - 2\alpha$. Then, as

$$\Pr\{h^{-1}(L) \leq \psi_0 \leq h^{-1}(U)\} \doteq 1 - 2\alpha,$$

the interval $(h^{-1}(L), h^{-1}(U))$ is a $(1 - 2\alpha)$ confidence interval for ψ_0 . Which of the many possible transformations h should we use? Sometimes the choice is suggested by the need to avoid intervals that contain silly values of ψ , as in the following example.

Example 3.7 (Binomial distribution) Suppose that we want a 95% confidence interval for the support π for a small political party, based on a sample of $m = 100$ individuals. If $\hat{\pi} = 0.02$, the standard error is $(0.02 \times 0.98/100)^{1/2} = 0.014$, so the 95% interval, roughly $(-0.008, 0.034)$, contains negative values of π .

To avoid this, let us construct an interval for $h(\pi) = \log \pi$ instead, so that $h'(\pi) = \pi^{-1}$. Now $\log \hat{\pi} = -3.91$, with standard error $\hat{\pi}^{-1}\{\hat{\pi}(1 - \hat{\pi})/m\}^{1/2} = 0.7$. Hence the 95% interval for $\log \pi$ is roughly $-3.91 \pm 1.96 \times 0.7$, and the corresponding interval for π is $(\exp(-3.91 - 1.4), \exp(3.91 + 1.4)) = (0.005, 0.08)$. The

Table 3.1 Exact mean and variance of variance-stabilized form $Y^{1/2}$ of Poisson random variable.

θ	0.25	0.5	1	2	5	10	20
$E(Y^{1/2})$	0.23	0.44	0.77	1.27	2.17	3.12	4.44
$\text{var}(Y^{1/2})$	0.20	0.31	0.40	0.39	0.29	0.26	0.26

distribution of R/m is too far from normal here to take this interval very seriously, but at least it contains only positive values. ■

A different approach is to choose a transformation for which $\text{var}\{h(T)\}$ is roughly constant, independent of ψ . Let T be an estimator of ψ , and suppose that $\text{var}(T) = \phi V(\psi)/n$, where ϕ is independent of ψ . The function $V(\psi)$ is called the *variance function* of T . We aim to choose h such that

$$1 \propto \text{var}\{h(T)\} \doteq h'(\psi)^2 \text{var}(T) = h'(\psi)^2 \phi V(\psi)/n,$$

where the approximation results from the delta method. This implies that

$$h(\psi) \propto \int^\psi \frac{du}{V(u)^{1/2}}, \quad (3.4)$$

which is called the *variance-stabilizing transformation* for T .

Example 3.8 (Poisson distribution) The mean and variance of the Poisson density (2.6) are both θ , so the average of a random sample of n such variables has mean θ and variance θ/n , giving $V(\theta) = \theta$ and $\phi = 1$. The variance-stabilizing transform is $h(\theta) = \int^\theta u^{-1/2} du \propto \theta^{1/2}$; the constant of proportionality is irrelevant. The delta method gives $\text{var}(Y^{1/2}) \doteq 0.25$. The exact mean and variance of $Y^{1/2}$ are given in Table 3.1. Variance-stabilization does not work perfectly, but $\text{var}(Y^{1/2})$ depends much less on θ than $\text{var}(Y)$ does.

To apply this to the birth data, we use the 16 arrivals on the first day. To construct a $(1 - 2\alpha)$ confidence interval for the mean arrivals per day, we recall that the Poisson mean and variance both equal θ and suppose that $(Y - \theta)/\theta^{1/2} \sim N(0, 1)$. An estimator of the denominator is $Y^{1/2}$, and taking $(Y - \theta)/Y^{1/2} \sim N(0, 1)$ gives $(Y - Y^{1/2}z_{1-\alpha}, Y - Y^{1/2}z_\alpha)$ as approximate confidence interval. With $\alpha = 0.025$ and $y = 16$ this yields (8.2, 23.8).

It is better to take $Y^{1/2} \sim N(\theta^{1/2}, 0.25)$, giving $(1 - 2\alpha)$ confidence intervals

$$\left(Y^{1/2} - \frac{1}{2}z_{1-\alpha}, Y^{1/2} - \frac{1}{2}z_\alpha\right), \quad \left(\left(Y^{1/2} - \frac{1}{2}z_{1-\alpha}\right)^2, \left(Y^{1/2} - \frac{1}{2}z_\alpha\right)^2\right)$$

for $\theta^{1/2}$ and θ . With $\alpha = 0.025$ and $y = 16$ this gives (9.1, 24.8), which is shifted to the right relative to the interval above, and is not symmetric about y . Here the effect of transformation is small, but it can be much larger in other problems. ■

3.1.3 Tests

The distribution of the pivot $Z(\psi_0)$ implies that some values of ψ are more plausible than others, and we can gauge this using confidence intervals: values of ψ close to the centre of a (say) 95% confidence interval are evidently more plausible than are those that only just lie within it. In some applications a particular value of ψ has special meaning and we may want to assess its plausibility in the light of some data. Given a set of data, a pivot $Z(\psi)$ and a value ψ_0 whose plausibility we wish to establish, one approach is to obtain the observed value of the pivot, $z(\psi_0)$, and then regard the probability $\Pr\{Z(\psi_0) \leq z(\psi_0)\}$ as a measure of the consistency of ψ_0 with the data. The key point is that if ψ_0 was the value of ψ which generated the data, then we would expect $z(\psi_0)$ to be a plausible value for $Z(\psi_0)$, but if not, we would expect $z(\psi_0)$ to be more extreme relative to the known distribution of the pivot.

Example 3.9 (Birth data) If the average time in the delivery suite for 10,000 women at a hospital in Manchester was 6 hours, then we might want to see if this is consistent with the times in Oxford; the Manchester sample is so large that we can treat the 6 hours as fixed. The times for day 1 of the Oxford data seem longer, but how sure can we be?

If ψ_0 for Oxford was equal to 6 hours, then the observed value of $Z(\psi_0)$ for day 1 of the Oxford data,

$$z(\psi_0) = (\bar{y} - \psi_0)/(s^2/n)^{1/2} = (8.77 - 6)/(18.46/16)^{1/2} = 2.58,$$

would be the value of an approximately normal variable. However this seems unlikely: with ψ_0 equal to 6 we get

$$\Pr\{Z(\psi_0) \leq 2.58\} \doteq \Phi(2.58) = 0.995.$$

This is an event which might take place about once in 200 repetitions, and it suggests two possibilities: either the Manchester and Oxford data actually are consistent but an unusual event has occurred, or they are not consistent, and in fact the average time is indeed shorter in Manchester. ■

Tests and their relation to confidence intervals are discussed further in Sections 4.5 and 7.3.4.

3.1.4 Prediction

In some applications the focus of interest is the likely value of an as-yet unobserved random variable Y_+ , to be predicted using known data y , taken to be a realization of a random variable Y . By analogy with using pivots to make inferences on unknown parameters, it may then be possible to construct a function $Q = q(Y_+, Y)$ whose distribution is independent of the parameters and such that

$$\Pr\{q(Y_+, Y) \in R_\alpha\} = \Pr\{l_\alpha(Y) \leq Y_+ \leq u_\alpha(Y)\} = 1 - 2\alpha.$$

Prediction intervals are also known as tolerance intervals.

Then $(l_\alpha(y), u_\alpha(y))$ is a $(1 - 2\alpha)$ prediction interval for Y_+ .

Example 3.10 (Location-scale model) Suppose that Y_+ is to be predicted using an independent random sample $Y_1, \dots, Y_n$ from a location-scale model. We can write $Y_+ = \eta + \tau \varepsilon_+$ and $Y_j = \eta + \tau \varepsilon_j$, where the ε s have common and known density g , say. If $\bar{Y}$ and S^2 are the sample average and variance of $Y_1, \dots, Y_n$, then the distribution of $Q = (Y_+ - \bar{Y})/S$ depends only on g , and its quantiles q_α may be found numerically. Then

$$\Pr\{q_\alpha \leq (Y_+ - \bar{Y})/S \leq q_{1-\alpha}\} = \Pr(\bar{Y} + Sq_\alpha \leq Y_+ \leq \bar{Y} + Sq_{1-\alpha}) = 1 - 2\alpha,$$

and hence $(\bar{y} + sq_\alpha, \bar{y} + sq_{1-\alpha})$ is an equitailed $(1 - 2\alpha)$ prediction interval for Y_+ . ■

Exercises 3.1

- Calculate a two-sided 0.95 confidence interval for the mean population time in the delivery suite based on day 2 of the data in Table 2.1. Obtain also lower and upper 0.90 confidence intervals.
- Let $Y_1, \dots, Y_n$ be defined by $Y_j = \mu + \sigma X_j$, where $X_1, \dots, X_n$ is a random sample from a known density g with distribution function G . If $M = m(Y)$ and $S = s(Y)$ are location and scale statistics based on $Y_1, \dots, Y_n$, that is, they have the properties that $m(Y) = \mu + \sigma m(X)$ and $s(Y) = \sigma s(X)$ for all $X_1, \dots, X_n$, $\sigma > 0$ and real μ , then show that $Z(\mu) = n^{1/2}(M - \mu)/S$ is a pivot.
When n is odd and large, g is the standard normal density, M is the median of $Y_1, \dots, Y_n$ and $S = \text{IQR}$ their interquartile range, show that $S/1.35 \xrightarrow{P} \sigma$, and hence show that as $n \rightarrow \infty$, $Z(\mu) \xrightarrow{D} N(0, \tau^2)$, for known $\tau > 0$. Hence give the form of a 95% confidence interval for μ .
Compare this interval and that based on using $Z(\mu)$ with $M = \bar{Y}$ and S^2 the sample variance, for the data for day 4 in Table 2.1.
- If Y is Poisson with large mean θ , then $(Y - \theta)/\theta^{1/2} \sim N(0, 1)$. Show that the limits of a $(1 - 2\alpha)$ confidence interval for θ are the solutions of the equation $(Y - \theta)^2 = z_\alpha^2 \theta$. Obtain them and compare them with the intervals for the birth data in Example 3.8.
- Suppose that the unemployment rate π is estimated by sampling randomly from the potential workforce. A total of m individuals are sampled and the number unemployed R is found, giving $\hat{\pi} = R/m$. How large should m be if $\pi \doteq 0.05$ and a standard error of at most 0.005 is required? What if $\pi = 0.1$?
In some countries such surveys are conducted by telephone interviews with a fixed number of households chosen randomly from the phone book and then asking how many people in the household are eligible for work (not children, retired, ...) and how many are working. Suppose that the total number of people is n , of whom M are eligible to work; suppose that M is binomial with denominator n and probability θ . Of the M , R are eligible to work, so $\hat{\pi} = R/M$ with M now random. If $n = 12,000$, $\theta = 0.5$ and $\pi = 0.05$, use the delta method to compute a variance for $\hat{\pi}$. Compute also the variance when $M = 6000$ is treated as fixed. Does the variability of M change the variance by much?
What problems might arise when sampling from the phone book?
- One way to construct a confidence interval for a real parameter θ is to take the interval $(-\infty, \infty)$ with probability $(1 - 2\alpha)$, and otherwise take the empty set $\emptyset$. Show that this procedure has exact coverage $(1 - 2\alpha)$. Is it a good procedure?
- A binomial variable R has mean $m\pi$ and variance $m\pi(1 - \pi)$. Find the variance function of $Y = R/m$, and hence obtain the variance-stabilizing transform for R .

- 7 Let I be a confidence interval for μ based on an estimator T whose distribution is $N(\mu, \sigma^2)$. Show that $\exp(I)$ is a confidence interval for the median of the distribution of $\exp(T)$. How would you compute a confidence interval for its mean, if σ^2 is (i) known and (ii) unknown?
- 8 If R is binomial with denominator m and probability π , show that

$$\frac{R/m - \pi}{\{\pi(1 - \pi)/m\}^{1/2}} \xrightarrow{D} Z \sim N(0, 1),$$

and that the limits of a $(1 - 2\alpha)$ confidence interval for π are the solutions to

$$R^2 - (2mR + mz_\alpha^2)\pi + m(m + z_\alpha^2)\pi^2 = 0.$$

Give expressions for them.

In a sample with $m = 100$ and 20 positive responses, the 0.95 confidence interval is (0.13, 0.29). As this interval either does or does not contain the true π , what is the meaning of the 0.95?

- 9 I am uncertain about what will happen when I next roll a die, about the exact amount of money at present in my bank account, about the weather tomorrow, and about what will happen when I die. Does uncertainty mean the same thing in all these contexts? For which is variation due to repeated sampling meaningful, do you think?
- 10 Let $Y_1, \dots, Y_n$ be a random sample from a model in which $Y_j = \theta X_j$, where the X_j are independent with known density g . Show that $\sum Y_j/\theta$ is a pivot, and deduce that a $(1 - 2\alpha)$ confidence interval for θ based on $\sum Y_j$ has form $(\sum Y_j/a, \sum Y_j/b)$, where a and b are known constants. If $g(x) = e^{-x}$, $x > 0$, is the exponential density, then the 0.025, 0.05, 0.1, 0.5, 0.9, 0.95 and 0.975 quantiles of $\sum X_j$ for $n = 12$ are 6.20, 6.92, 7.83, 11.67, 16.60, 18.21 and 19.68. Use them to give two-sided 0.80 and 0.95 confidence intervals for θ , based on the data in Practical 2.5. Give also upper and lower 0.90 confidence intervals for θ .

3.2 Normal Model

3.2.1 Normal and related distributions

The previous section described an approach to approximate statements of uncertainty, useful in many contexts. We now discuss exact inference for a model of central importance, when the data available form a random sample from the normal distribution. That is, we treat the data $y_1, \dots, y_n$ as the observed values of $Y_1, \dots, Y_n$, where the Y_j are independently taken from the normal density

$$f(y; \mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp \left\{ -\frac{1}{2\sigma^2}(y - \mu)^2 \right\}, \quad -\infty < y < \infty, \quad (3.5)$$

with μ real and σ positive. The normal model owes its ubiquity to the central limit theorem, which, in addition to applying to functions of many observations, may apply to individual measurements themselves. For example, in Example 1.1 it is reasonable to suppose that a plant's height is determined by the effects of many genes, to which an averaging effect may apply, leading to a normal distribution of heights for the population to which the individual belongs, and therefore suggesting the use of normal distributions in (1.1), (1.2), and (1.3). In other situations the simplicity of inference for the normal distribution leads to its use as an approximation even where no such

Laplace named this the Gaussian density, after Johann Carl Friedrich Gauss (1777–1855), who derived it while writing on the combination of astronomical observations by least squares.

argument applies. Of course it is important to check that the data do appear normally distributed, for example by a normal probability plot (Section 2.1.4).

Before considering inference for the normal sample, we discuss the normal and some related distributions. All are widely tabulated, and their density and distribution functions and quantiles are readily calculated in statistical packages.

Normal distribution

If we change variable in (3.5) from y to $z = (y - \mu)/\sigma$, we see that the corresponding random variable $Z = (Y - \mu)/\sigma$ has density

$$\phi(z) = (2\pi)^{-1/2} \exp\left(-\frac{1}{2}z^2\right), \quad -\infty < z < \infty; \quad (3.6)$$

this is the density of the *standard normal* random variable Z . The density (3.6) is symmetric about $z = 0$, and $E(Z) = 0$ and $\text{var}(Z) = 1$ (Exercise 3.2.1). Consequently the mean and variance of $Y = \mu + \sigma Z$ are μ and σ^2 . We write $Y \sim N(\mu, \sigma^2)$ as shorthand for ‘ Y has the normal distribution with mean μ and variance σ^2 ’.

The distribution function corresponding to (3.6),

$$\Phi(z) = (2\pi)^{-1/2} \int_{-\infty}^z \exp\left(-\frac{1}{2}u^2\right) du, \quad (3.7)$$

has no closed form, and neither do its quantiles, $z_p = \Phi^{-1}(p)$. Two useful values are $z_{0.025} = -1.96$ and $z_{0.05} = -1.65$. The symmetry of (3.6) about $z = 0$ implies that $z_p = -z_{1-p}$.

The moment-generating function of Y is

$$\begin{aligned} M(t) &= E(e^{tY}) \\ &= \frac{1}{(2\pi\sigma^2)^{1/2}} \int_{-\infty}^{\infty} \exp\left\{ty - \frac{1}{2\sigma^2}(y - \mu)^2\right\} dy \\ &= \frac{1}{(2\pi\sigma^2)^{1/2}} \int_{-\infty}^{\infty} \exp\left\{\mu t + \sigma^2 \frac{t^2}{2} - \frac{1}{2\sigma^2}(y - \mu - t\sigma)^2\right\} dy \\ &= \exp(\mu t + \sigma^2 t^2/2) \int_{-\infty}^{\infty} f(y; \mu + \sigma t, \sigma^2) dy \\ &= \exp(\mu t + \sigma^2 t^2/2), \end{aligned} \quad (3.8)$$

since for any real t , $f(y; \mu + \sigma t, \sigma^2)$ is just a normal density and has unit integral. We often use variants of this argument to sidestep integration.

The mean and variance of Y can be read off from its cumulant-generating function, $K(t) = \log M(t) = \mu t + \sigma^2 t^2/2$: $\kappa_1 = E(Y) = \mu$ and $\kappa_2 = \text{var}(Y) = \sigma^2$.

Chi-squared distribution

If $Z_1, \dots, Z_v$ are independent standard normal random variables, we say that $W = Z_1^2 + \dots + Z_v^2$ has the *chi-squared distribution on v degrees of freedom*: we write $W \sim \chi_v^2$. The probability density function of W ,

$$f(w) = \frac{1}{2^{v/2}\Gamma(v/2)} w^{v/2-1} e^{-w/2}, \quad w > 0, \quad v = 1, 2, \dots, \quad (3.9)$$

See Lindley and Scott (1984) or Pearson and Hartley (1976), for example.

Here $\Gamma(\kappa) = \int_0^\infty u^{\kappa-1} e^{-u} du$ is the gamma function; see Exercise 2.1.3.

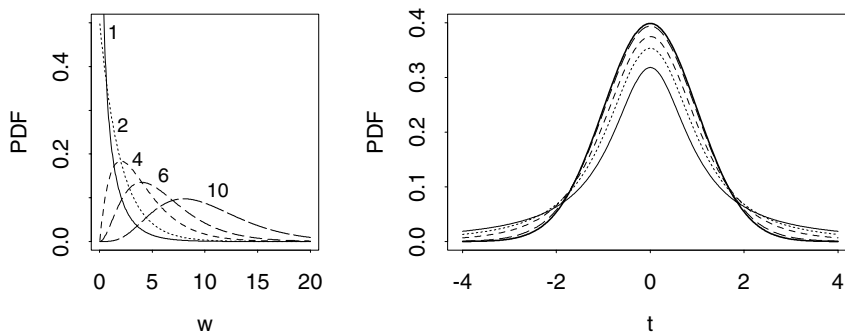


Figure 3.3 Chi-squared and Student t density functions (3.9) and (3.11). Left panel: chi-squared densities with 1 (solid), 2 (dots), 4 (dashes), 6 (larger dashes), and 10 (largest dashes) degrees of freedom. Right panel: t densities with 1 (solid), 2 (dots), 4 (dashes), and 20 (large dashes) degrees of freedom, and standard normal density (heavy solid). The scale is chosen to show the much heavier tails of the t density with few degrees of freedom.

is shown in the left panel of Figure 3.3 for various values of ν . As one would expect from its definition, both the mean and variance of W increase with ν . Its p quantile, denoted $c_\nu(p)$, has the property that $\Pr\{W \leq c_\nu(p)\} = p$. When $\nu = 1$, $W = Z^2$, where $Z \sim N(0, 1)$, so

$$\Pr(W \leq w) = \Pr(-\sqrt{w} \leq Z \leq \sqrt{w}),$$

implying that $c_1(1 - 2p) = z_p^2$.

It is clear from the definition of W that if $W_1 \sim \chi_{\nu_1}^2$ and $W_2 \sim \chi_{\nu_2}^2$ and they are independent, then $W_1 + W_2 \sim \chi_{\nu_1 + \nu_2}^2$; evidently this extends to finite sums of independent chi-squared variables. Chi-squared and gamma distributions are closely related: if X has the gamma density (2.7) with parameter λ and shape κ , then $\lambda X \sim \frac{1}{2} \chi_{2\kappa}^2$ (Exercise 3.2.2).

To find the moment-generating function of W , we first find the moment-generating function of Z_j^2 , namely

$$\begin{aligned} E(e^{tZ_j^2}) &= \frac{1}{(2\pi)^{1/2}} \int_{-\infty}^{\infty} e^{tz^2 - z^2/2} dz \\ &= (1 - 2t)^{-1/2} \frac{1}{(2\pi)^{1/2}} \int_{-\infty}^{\infty} e^{-u^2/2} du \\ &= (1 - 2t)^{-1/2}, \quad t < \frac{1}{2}, \end{aligned} \quad (3.10)$$

where we have changed variable from z to $u = (1 - 2t)^{1/2}z$. The Z_j^2 are independent and identically distributed, so W has moment-generating function $\{(1 - 2t)^{-1/2}\}^\nu = (1 - 2t)^{-\nu/2}$, differentiation of which shows that the mean and variance of W are ν and 2ν .

Student t distribution

Suppose now that Z and W are independent, that Z is standard normal and W is chi-squared with ν degrees of freedom, and let $T = Z/(W/\nu)^{1/2}$. The random variable T is said to have a *Student t distribution on ν degrees of freedom*; we write $T \sim t_\nu$. Its density is

$$f(t) = \frac{\Gamma\{(\nu + 1)/2\}}{\sqrt{\nu\pi}\Gamma(\nu/2)} \frac{1}{(1 + t^2/\nu)^{(\nu+1)/2}}, \quad -\infty < t < \infty, \quad \nu = 1, 2, \dots \quad (3.11)$$

The right panel of Figure 3.3 shows (3.11) for various values of ν . The distribution of T approaches that of Z for large ν , because the fact that $W/\nu \xrightarrow{P} 1$ as $\nu \rightarrow \infty$ implies that $T \xrightarrow{D} Z$; see Example 2.22. The extra variability induced by dividing Z by $(W/\nu)^{1/2}$ spreads out the distribution of T relative to that of Z , by a large amount when ν is small, but by less when ν is large. One consequence of this is that as $\nu \rightarrow \infty$ the quantiles of T , denoted $t_\nu(p)$, approach those of Z , that is, $t_\nu(p) \rightarrow z_p$. For example, the 0.025 quantiles for $\nu = 2, 10$, and 20 are $-4.30, -2.23$ and -2.09 , while $t_\infty(0.025) = z_{0.025} = -1.96$. The symmetry of (3.11) about $t = 0$ implies that $t_\nu(p) = -t_\nu(1 - p)$.

Not all the moments of T are finite, because the function $t^r f(t)$ is integrable only if $r < \nu$. One simple way to calculate its mean and variance, when they exist, is to use the identities

$$E\{h(Z, W)\} = E_W[E\{h(Z, W) \mid W\}], \quad (3.12)$$

$$\text{var}\{h(Z, W)\} = E_W[\text{var}\{h(Z, W) \mid W\}] + \text{var}_W[E\{h(Z, W) \mid W\}], \quad (3.13)$$

which hold for any random variables Z and W ; the inner expectation and variance are over the distribution of Z for W fixed (Exercise 3.2.3). If $h(Z, W) = Z/(W/\nu)^{1/2}$ and Z and W are independent, then

$$\begin{aligned} E\{Z/(W/\nu)^{1/2} \mid W\} &= (W/\nu)^{-1/2}E(Z) = 0, \\ \text{var}\{Z/(W/\nu)^{1/2} \mid W\} &= (W/\nu)^{-1}\text{var}(Z) = (W/\nu)^{-1}. \end{aligned}$$

Consequently (3.12) and (3.13) imply that $E(T) = E_W\{Z/(W/\nu)^{1/2}\} = 0$ and

$$\begin{aligned} \text{var}(T) &= E_W(\nu/W) \\ &= \frac{\nu}{2^{v/2}\Gamma(v/2)} \int_0^\infty w^{-1} \cdot w^{v/2-1} e^{-w/2} dw \\ &= \frac{\nu}{2^{v/2}\Gamma(v/2)} 2^{v/2-1} \Gamma(v/2 - 1) \\ &= \frac{\nu}{\nu - 2}, \quad \nu = 3, 4, \dots, \end{aligned}$$

the first equality following from (3.13), the second from (3.9), the third on noticing that the integrand is proportional to the chi-squared density on $\nu - 2$ degrees of freedom — whose integral must equal one — and the fourth on using the fact that $\Gamma(\kappa + 1) = \kappa \Gamma(\kappa)$, for $\kappa > 0$ (Exercise 2.1.3). The variance of T is finite only if $\nu \geq 3$, and its mean is finite only if $\nu \geq 2$. Setting $\nu = 1$ in (3.11) gives the Cauchy density (2.16), useful for counter-examples.

F distribution

Suppose that W_1 and W_2 have independent chi-squared distributions with ν_1 and ν_2 degrees of freedom respectively. Then

$$F = \frac{W_1/\nu_1}{W_2/\nu_2}$$

has the F distribution on v_1 and v_2 degrees of freedom: we write $F \sim F_{v_1, v_2}$. Its density function is

$$f(u) = \frac{\Gamma\left(\frac{1}{2}v_1 + \frac{1}{2}v_2\right) v_1^{v_1/2} v_2^{v_2/2}}{\Gamma\left(\frac{1}{2}v_1\right) \Gamma\left(\frac{1}{2}v_2\right)} \frac{u^{\frac{1}{2}v_1-1}}{(v_2 + v_1 u)^{(v_1+v_2)/2}}, \quad u > 0, \quad v_1, v_2 = 1, 2, \dots, \quad (3.14)$$

and its p quantile is denoted $F_{v_1, v_2}(p)$. When $v_1 = 1$, $F = Z^2/(W_2/v_2)$, where $Z \sim N(0, 1)$ is independent of $W_2 \sim \chi_{v_2}^2$, so F then has the same distribution as T^2 , where $T \sim t_{v_2}$.

3.2.2 Normal random sample

When a random sample $Y_1, \dots, Y_n$ is normal, there are compelling reasons to base inference for μ and σ^2 on its average and variance, $\bar{Y}$ and S^2 . At the end of this section we shall prove that their joint distribution is given by

We suppose that n is two or more, so $S^2 > 0$ with probability 1.

$$\left. \begin{aligned} \bar{Y} &\sim N(\mu, n^{-1}\sigma^2), \\ (n-1)S^2 &\sim \sigma^2 \chi_{n-1}^2, \end{aligned} \right\} \text{ independently.} \quad (3.15)$$

Another way to express this is

$$\left. \begin{aligned} \bar{Y} &\stackrel{D}{=} \mu + n^{-1/2}\sigma Z, & Z &\sim N(0, 1), \\ S^2 &\stackrel{D}{=} (n-1)^{-1}\sigma^2 W, & W &\sim \chi_{n-1}^2, \end{aligned} \right\} Z, W \text{ independent.}$$

The studentized form of $\bar{Y}$ may therefore be written

$$\begin{aligned} T &= \frac{\bar{Y} - \mu}{(S^2/n)^{1/2}} \\ &\stackrel{D}{=} \frac{n^{-1/2}\sigma Z}{\{\sigma^2(n-1)^{-1}W/n\}^{1/2}} \\ &= \frac{Z}{\{W/(n-1)\}^{1/2}}, \end{aligned} \quad (3.16)$$

which has the t distribution with $n-1$ degrees of freedom.

As the distribution of $T = (\bar{Y} - \mu)/(S^2/n)^{1/2}$ is known, T is an exact pivot, and there is no need for large-sample approximation when a confidence interval is required for μ . That is,

$$\begin{aligned} 1 - 2\alpha &= \Pr \left\{ t_{n-1}(\alpha) \leq \frac{\bar{Y} - \mu}{(S^2/n)^{1/2}} \leq t_{n-1}(1 - \alpha) \right\} \\ &= \Pr \left\{ \bar{Y} - n^{-1/2}St_{n-1}(1 - \alpha) \leq \mu \leq \bar{Y} - n^{-1/2}St_{n-1}(\alpha) \right\}. \end{aligned}$$

As the t distribution is symmetric, the random interval with endpoints

$$\bar{Y} \pm n^{-1/2}St_{n-1}(\alpha) \quad (3.17)$$

contains μ with probability exactly $(1 - 2\alpha)$, for all $n \geq 2$. In practice, $\bar{Y}$ and S are replaced by their observed values $\bar{y}$ and s , and the resulting interval has the repeated sampling interpretation outlined in Section 3.1.

Example 3.11 (Maize data) The final column of Table 1.1 contains the differences in heights between $n = 15$ pairs of self- and cross-fertilized plants. Suppose that these differences are a random sample from the $N(\mu, \sigma^2)$ distribution; here μ and σ have units of eighths of an inch, and represent the mean and standard deviation of a population of such differences.

The values of the average and sample variance are $\bar{y} = 20.93$ and $s^2 = 1424.6$. As $t_{14}(0.025) = -2.14$, the 95% confidence interval for μ is $\bar{y} \pm n^{-1/2}st_{n-1}(\alpha)$, that is, $20.93 \pm (1424.6/15)^{1/2} \times 2.14 = (0.03, 41.84)$ eighths of an inch. This interval suggests that the mean difference in heights is positive; the best estimate of μ is about $2\frac{1}{2}$ inches. However, the value $\mu = 0$ is only just outside the interval, so the evidence for a height difference between the two types of plants is not overwhelming. ■

A similar argument gives confidence intervals for σ^2 . If $(n-1)S^2 \sim \sigma^2\chi_{n-1}^2$, then $(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$ is another exact pivot. Thus

$$\Pr \left\{ c_{n-1}(\alpha) \leq \frac{(n-1)S^2}{\sigma^2} \leq c_{n-1}(1-\alpha) \right\} = 1 - 2\alpha,$$

leading to the exact $(1 - 2\alpha)$ confidence interval for σ^2 ,

$$((n-1)S^2/c_{n-1}(1-\alpha), (n-1)S^2/c_{n-1}(\alpha)). \quad (3.18)$$

Example 3.12 (Maize data) Table 1.1 shows samples of sizes $n_1 = n_2 = 15$ on the heights of plants; the sample variances are $s_1^2 = 837.3$ and $s_2^2 = 269.4$ for the cross- and self-fertilized plants respectively.

If we take $\alpha = 0.025$, then $c_{14}(0.025) = 5.629$ and $c_{14}(0.975) = 26.119$. Hence the 95% confidence interval (3.18) for the variance for the cross-fertilized data is $(14s_1^2/c_{14}(0.975), 14s_1^2/c_{14}(0.025))$, that is, $(449, 2082)$ eighths of inches squared. ■

The F distribution gives a means to compare the variances of two normal samples. Suppose that S_1^2 and S_2^2 are the sample variances for two independent normal samples of respective sizes n_1 and n_2 , and that the variances of those samples are σ^2 and $\psi\sigma^2$. That is, ψ is the ratio of the variances of the samples. Then $(n_1 - 1)S_1^2/\sigma^2$ and $(n_2 - 1)S_2^2/(\psi\sigma^2)$ have independent chi-squared distributions on $n_1 - 1$ and $n_2 - 1$ degrees of freedom, and

$$\Pr \left\{ F_{n_1-1, n_2-1}(\alpha) \leq \frac{S_1^2/\sigma^2}{S_2^2/(\psi\sigma^2)} \leq F_{n_1-1, n_2-1}(1-\alpha) \right\} = 1 - 2\alpha,$$

or equivalently

$$\Pr \left\{ F_{n_1-1, n_2-1}(\alpha) \frac{S_2^2}{S_1^2} \leq \psi \leq F_{n_1-1, n_2-1}(1-\alpha) \frac{S_2^2}{S_1^2} \right\} = 1 - 2\alpha.$$

Thus, given two normal random samples whose variances are s_1^2 and s_2^2 ,

$$(F_{n_1-1, n_2-1}(\alpha)s_2^2/s_1^2, F_{n_1-1, n_2-1}(1-\alpha)s_2^2/s_1^2) \quad (3.19)$$

is a $(1 - 2\alpha)$ confidence interval for the ratio of variances, ψ . Here the pivot is $\psi S_1^2/S_2^2$, which has an exact F_{n_1-1, n_2-1} distribution.

Example 3.13 (Maize data) Following on from Example 3.12, we take $\alpha = 0.025$, giving $F_{14,14}(0.025) = 0.336$, $F_{14,14}(0.975) = 2.979$. The 95% confidence interval (3.19) for the ratio of the variances for self- and cross-fertilized plants is (0.108, 0.958). The value $\psi = 1$ is not in this interval, which suggests that the self-fertilized plants are less variable in height than the cross-fertilized ones. ■

The comparison of variance estimates using F statistics is a crucial ingredient in the analysis of variance, discussed in Section 8.5.

3.2.3 Multivariate normal distribution

The normal distribution plays a central role in inference for scalar data. Its simple properties generalize elegantly to vectors of variables, and these we study now.

One measure of the strength of association between scalar random variables Y_1 and Y_2 is their *covariance*,

$$\text{cov}(Y_1, Y_2) = E[\{Y_1 - E(Y_1)\} \{Y_2 - E(Y_2)\}].$$

Evidently $\text{cov}(Y_1, Y_1) = \text{var}(Y_1)$, $\text{cov}(Y_1, Y_2) = \text{cov}(Y_2, Y_1)$, and if a and b are constants then $\text{cov}(a + bY_1, Y_2) = b\text{cov}(Y_1, Y_2)$.

In general we may have several random variables. If Y denotes the $p \times 1$ vector $(Y_1, \dots, Y_p)^T$ and Z denotes the $q \times 1$ vector $(Z_1, \dots, Z_q)^T$, let $E(Y)$ be the $p \times 1$ vector whose r th element is $E(Y_r)$. We define the covariance of Y and Z to be the $p \times q$ matrix

$$\text{cov}(Y, Z) = E[\{Y - E(Y)\} \{Z - E(Z)\}^T]$$

whose (r, s) element is $\text{cov}(Y_r, Z_s)$. In particular, $\text{cov}(Y, Y) = \Omega$, the $p \times p$ symmetric matrix whose (r, s) element is $\omega_{rs} = \text{cov}(Y_r, Y_s)$; this is called the *covariance matrix* of Y . It is symmetric because $\text{cov}(Y_r, Y_s) = \text{cov}(Y_s, Y_r)$, positive semi-definite because

Or sometimes just the *variance matrix*.

$$\text{var}(a^T Y) = \text{cov}(a^T Y, a^T Y) = a^T \text{cov}(Y, Y) a = a^T \Omega a \geq 0$$

for any constant $p \times 1$ vector a , and positive definite unless the distribution of Y is *degenerate*, here meaning that some Y_r is constant or can be expressed in terms of a linear combination of the others (Exercise 3.2.14).

The covariance matrix of the linear combinations $a + B^T Y$ and $c + D^T Y$, where a and c are respectively $q \times 1$ and $r \times 1$ constant vectors, and B and D are respectively $p \times q$ and $p \times r$ constant matrices, is

$$\begin{aligned} \text{cov}(a + B^T Y, c + D^T Y) &= E[\{B^T Y - E(B^T Y)\} \{D^T Y - E(D^T Y)\}^T] \\ &= E[B^T \{Y - E(Y)\} \{Y - E(Y)\}^T D] \\ &= B^T \Omega D. \end{aligned}$$

When a, b, c, d are constants, $\text{cov}(a + bY_1, c + dY_2) = bdcov(Y_1, Y_2)$, and thus covariance is not an absolute measure of the association between the variables, because it depends on their units. A measure that is invariant to the choice of units is the *correlation* of Y_1 and Y_2 , namely

$$\text{corr}(Y_1, Y_2) = \frac{\text{cov}(Y_1, Y_2)}{\{\text{var}(Y_1)\text{var}(Y_2)\}^{1/2}},$$

some of whose properties were outlined in Example 2.21 and Exercise 2.2.3. Positive correlation between Y_1 and Y_2 indicates that large values of Y_1 and Y_2 tend to occur together, and conversely; whereas negative correlation means that if Y_1 is larger than $E(Y_1)$, Y_2 tends to be smaller than $E(Y_2)$. The correlation matrix of a $p \times 1$ vector Y has as its (r, s) element the correlation between Y_r and Y_s , and may be expressed as $\Omega_d^{-1/2} \Omega \Omega_d^{-1/2}$, where Ω_d is the diagonal matrix $\text{diag}(\omega_{11}, \dots, \omega_{pp})$. The diagonal of $\Omega_d^{-1/2} \Omega \Omega_d^{-1/2}$ consists of ones.

Multivariate normal distribution

A p -dimensional *multivariate normal* random variable $Y = (Y_1, \dots, Y_p)^T$ with $p \times 1$ vector mean μ and $p \times p$ covariance matrix Ω has density

$$f(y; \mu, \Omega) = \frac{1}{(2\pi)^{p/2} |\Omega|^{1/2}} \exp \left\{ -\frac{1}{2} (y - \mu)^T \Omega^{-1} (y - \mu) \right\}; \quad (3.20)$$

we write $Y \sim N_p(\mu, \Omega)$. Here Y , y , and μ take values in $\mathbb{R}^p$. We assume that the distribution is not degenerate, in which case Ω is positive definite, implying amongst other things that its determinant $|\Omega| > 0$.

The moment-generating function of Y is

$$M(t) = E(e^{t^T Y}) = \frac{1}{(2\pi)^{p/2} |\Omega|^{1/2}} \int \exp \left\{ t^T y - \frac{1}{2} (y - \mu)^T \Omega^{-1} (y - \mu) \right\} dy,$$

where t^T is the $1 \times p$ vector $(t_1, \dots, t_p)$ and $Y = (Y_1, \dots, Y_p)^T$; the integral is over $y \in \mathbb{R}^p$. To simplify $M(t)$ we write the exponent inside the integral as

$$t^T \mu + \frac{1}{2} t^T \Omega t - \frac{1}{2} (y - \mu - \Omega t)^T \Omega^{-1} (y - \mu - \Omega t).$$

The first two terms of this do not depend on y , so

$$M(t) = \exp \left(t^T \mu + \frac{1}{2} t^T \Omega t \right) \int f(y; \mu + \Omega t, \Omega) dy = \exp \left(t^T \mu + \frac{1}{2} t^T \Omega t \right),$$

because for any value of μ , (3.20) is a probability density function. We obtain the moments of Y by differentiation:

$$\begin{aligned} E(Y_r) &= \frac{\partial M(0)}{\partial t_r} = \mu_r, \\ \text{cov}(Y_r, Y_s) &= \frac{\partial^2 M(0)}{\partial t_r \partial t_s} - \frac{\partial M(0)}{\partial t_r} \frac{\partial M(0)}{\partial t_s} = \omega_{rs} + \mu_r \mu_s - \mu_r \mu_s = \omega_{rs}. \end{aligned}$$

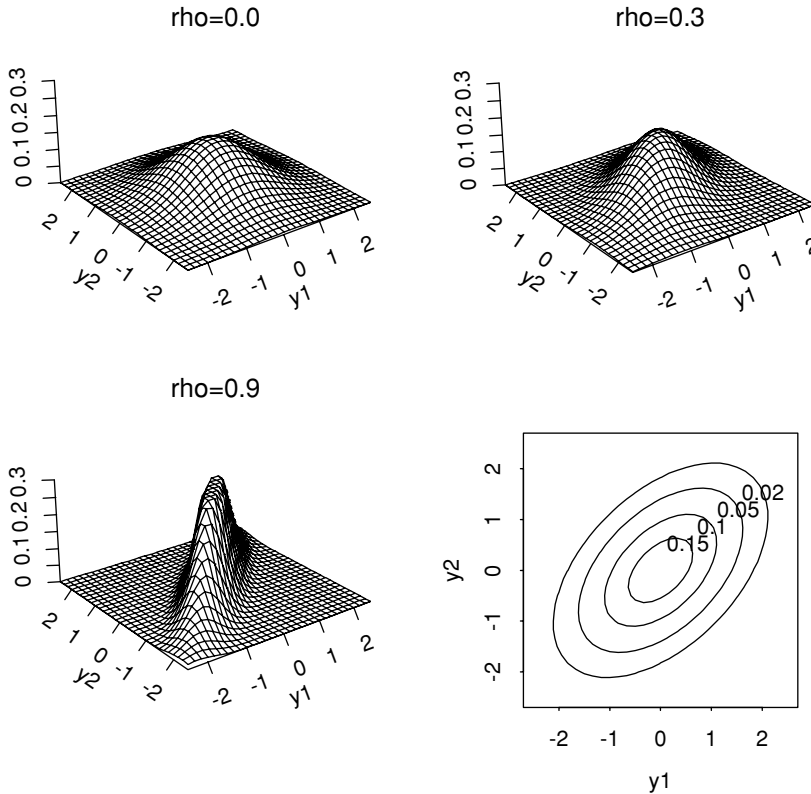


Figure 3.4 The bivariate normal density, with correlation $\rho = 0, 0.3$, and 0.9 . The lower right panel shows contours of the density when $\rho = 0.3$; note that they are elliptical. In higher dimensions the contours of equal density are ellipsoids.

The cumulant-generating function of Y is

$$K(t) = \log M(t) = t^T \mu + \frac{1}{2} t^T \Omega t = \sum_{r=1}^p t_r \mu_r + \frac{1}{2} \sum_{r=1}^p \sum_{s=1}^p t_r t_s \omega_{rs}.$$

Thus the first and second cumulants are $\kappa_r = \mu_r$ and $\kappa_{r,s} = \omega_{rs}$, which are respectively the r th element of μ and the (r, s) element of Ω ; all higher cumulants are zero.

A special case of (3.20) is the bivariate normal distribution, whose covariance matrix is

$$\begin{pmatrix} \omega_{11} & \omega_{12} \\ \omega_{21} & \omega_{22} \end{pmatrix};$$

the correlation between Y_1 and Y_2 is $\rho = \omega_{12}/(\omega_{11}\omega_{22})^{1/2}$. This density is shown in Figure 3.4 for $\mu = 0$; the effect of increasing ρ is to concentrate the probability mass close to the line $y_1 = y_2$. The corresponding densities for negative ρ are obtained by reflection in the line $y_1 = 0$. When $p = 2$ the contours of constant density are ellipses, but when $p > 2$ they are the ellipsoids given by constant values of $(y - \mu)^T \Omega^{-1} (y - \mu)$.

Marginal and conditional distributions

To study the distribution of a subset of Y , we write $Y^T = (Y_1^T, Y_2^T)$, where now Y_1 has dimension $q \times 1$ and Y_2 has dimension $(p - q) \times 1$. Partition t , μ , and Ω conformably, so that

$$t = \begin{pmatrix} t_1 \\ t_2 \end{pmatrix}, \quad \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \Omega = \begin{pmatrix} \Omega_{11} & \Omega_{12} \\ \Omega_{21} & \Omega_{22} \end{pmatrix},$$

where t_1 and μ_1 are $q \times 1$ vectors and Ω_{11} is a $q \times q$ matrix, t_2 and μ_2 are $(p - q) \times 1$ vectors and Ω_{22} is a $(p - q) \times (p - q)$ matrix, and $\Omega_{12} = \Omega_{21}^T$ is a $q \times (p - q)$ matrix. The moment-generating function of Y is

$$\begin{aligned} E(e^{t^T Y}) &= E(e^{t_1^T Y_1 + t_2^T Y_2}) \\ &= \exp \left\{ t_1^T \mu_1 + t_2^T \mu_2 + \frac{1}{2} (t_1^T \Omega_{11} t_1 + 2t_1^T \Omega_{12} t_2 + t_2^T \Omega_{22} t_2) \right\}, \end{aligned}$$

from which we obtain the moment-generating functions of Y_1 and Y_2 by setting t_2 and t_1 respectively equal to zero, giving

$$E(e^{t_1^T Y_1}) = \exp(t_1^T \mu_1 + \frac{1}{2} t_1^T \Omega_{11} t_1), \quad E(e^{t_2^T Y_2}) = \exp(t_2^T \mu_2 + \frac{1}{2} t_2^T \Omega_{22} t_2).$$

Thus the marginal distributions of Y_1 and Y_2 are multivariate normal also. Note that Y_1 and Y_2 are independent if and only if their joint moment-generating function factorizes, that is,

$$E(e^{t_1^T Y_1 + t_2^T Y_2}) = E(e^{t_1^T Y_1}) E(e^{t_2^T Y_2}), \quad \text{for all } t_1, t_2,$$

which occurs if and only if $\Omega_{12} = \Omega_{21}^T = 0$.

Equivalently and more elegantly, the cumulant-generating function of Y_1 and Y_2 is

$$K(t_1, t_2) = t_1^T \mu_1 + t_2^T \mu_2 + \frac{1}{2} (t_1^T \Omega_{11} t_1 + 2t_1^T \Omega_{12} t_2 + t_2^T \Omega_{22} t_2),$$

and Y_1 and Y_2 are independent if and only if its coefficient in t_1 and t_2 , $t_1^T \Omega_{12} t_2$, is identically zero; this is the case if $\Omega_{12} = 0$ but not otherwise.

Thus for normal random variables zero covariance is equivalent to independence. One implication is that if $Y_1, \dots, Y_n$ is a random sample from the normal distribution with mean μ and variance σ^2 , then we can write

$$Y \sim N_n(\mu 1_n, \sigma^2 I_n).$$

The conditional distribution of Y_1 given that $Y_2 = y_2$ is (Exercise 3.2.18)

$$N_q(\mu_1 + \Omega_{12} \Omega_{22}^{-1} (y_2 - \mu_2), \Omega_{11} - \Omega_{12} \Omega_{22}^{-1} \Omega_{21}). \quad (3.21)$$

In the bivariate normal distribution with zero mean and unit variances,

$$N_2 \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right\},$$

the conditional mean of Y_1 given $Y_2 = y_2$ is ρy_2 , and the conditional variance is $1 - \rho^2$. Thus $\text{var}(Y_1 | Y_2 = y_2) \rightarrow 0$ as $|\rho| \rightarrow 1$. In the lower right panel of Figure 3.4 this

1_n denotes the $n \times 1$ vector of 1s and I_n the $n \times n$ identity matrix.

conditional density is supported on a horizontal line passing through y_2 , and the conditional mean of Y_1 increases with y_2 .

Example 3.14 (Trivariate distribution) Let $Y \sim N_3(\mu, \Omega)$, where

$$\mu = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad \Omega = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

The marginal distribution of Y_1 is $N(1, 2)$ and the marginal distribution of $(Y_1, Y_2)^T$ is

$$N_2 \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \right\};$$

Y_1 and Y_2 are marginally independent.

For the conditional distribution of $(Y_1, Y_2)^T$ given Y_3 we set

$$\mu_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mu_2 = (1), \quad \Omega_{11} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad \Omega_{12} = \Omega_{21}^T = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \Omega_{22} = (2).$$

Given $Y_3 = y_3$, $(Y_1, Y_2)^T$ is bivariate normal with mean vector and variance matrix

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} 2^{-1}(y_3 - 1), \quad \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} 2^{-1}(1, 1) = \begin{pmatrix} 3/2 & -1/2 \\ -1/2 & 3/2 \end{pmatrix}.$$

Thus knowledge of Y_3 induces correlation between Y_1 and Y_2 despite their marginal independence. Moreover the conditional variance of Y_1 is smaller than the marginal variance: knowing Y_3 makes one more certain about Y_1 . The positive covariance between Y_1 and Y_3 means that if Y_3 is known to exceed its mean, that is, $y_3 > 1$, then the conditional mean of Y_1 exceeds its marginal mean by an amount that depends on the difference $y_3 - 1$. ■

Linear combinations of normal variables

Linear combinations of normal random variables often arise. The moment-generating function of the linear combination $a + b^T Y$, where the constants a and b are respectively a scalar and a $p \times 1$ vector, is

$$\begin{aligned} E\{e^{t(a+b^T Y)}\} &= e^{ta} \exp \left\{ (bt)^T \mu + \frac{1}{2} (bt)^T \Omega (bt) \right\} \\ &= \exp \left\{ t(a + b^T \mu) + \frac{t^2}{2} b^T \Omega b \right\}, \end{aligned}$$

and hence $a + b^T Y$ has the normal distribution with mean $a + b^T \mu$ and variance $b^T \Omega b$. This extends to vectors $U = a + B^T Y$, where a is a $q \times 1$ constant vector and B is a $p \times q$ constant matrix. Then U has moment-generating function

$$\begin{aligned} E(e^{t^T U}) &= e^{t^T a} E(e^{t^T B^T Y}) = e^{t^T a} E(e^{(Bt)^T Y}) \\ &= \exp \left\{ t^T a + (Bt)^T \mu + \frac{1}{2} (Bt)^T \Omega (Bt) \right\} \\ &= \exp \left\{ t^T (a + B^T \mu) + \frac{1}{2} t^T B^T \Omega B t \right\}, \end{aligned}$$

and so U has a multivariate normal distribution with $q \times 1$ mean $a + B^T\mu$ and $q \times q$ covariance matrix $B^T\Omega B$; this is singular and the distribution degenerate unless B has full rank and $q \leq p$. That is, if $Y \sim N_p(\mu, \Omega)$, then

$$a + B^TY \sim N_q(a + B^T\mu, B^T\Omega B). \quad (3.22)$$

Example 3.15 (Trivariate distribution) In the previous example, consider the joint distribution of $U_1 = Y_1 + Y_2 + Y_3 - 4$ and $U_2 = Y_1 - Y_2 + Y_3$:

$$U = \begin{pmatrix} -4 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix}.$$

The mean vector and covariance matrix of U are

$$\begin{aligned} & \begin{pmatrix} -4 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \\ & \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 10 & 4 \\ 4 & 6 \end{pmatrix}. \end{aligned}$$

■

A further consequence of (3.22) follows from the spectral decomposition $\Omega = ELE^T$, where the columns of E are eigenvectors of Ω , L is the diagonal matrix containing the corresponding eigenvalues, and $EE^T = E^TE = I_p$. For positive definite Ω , the elements of L are strictly positive and hence $\Omega^{-1} = EL^{-1}E^T$. We set $U = L^{-1/2}E^T(Y - \mu)$, and note that $U \sim N_p(0, I_p)$, so

$$(Y - \mu)^T\Omega^{-1}(Y - \mu) = (Y - \mu)^TE L^{-1}E^T(Y - \mu) = U^TU \sim \chi_p^2. \quad (3.23)$$

Two samples

Result (3.22) has many uses. For example, suppose that a random sample of size n_1 is available from the $N(\mu_1, \sigma_1^2)$ density and an independent random sample of size n_2 is available from the $N(\mu_2, \sigma_2^2)$ density, and that the focus of interest is the difference of means $\mu_1 - \mu_2$. This is the situation in Example 1.1. Then since (3.15) applies to each sample separately,

$$\begin{pmatrix} \bar{Y}_1 \\ \bar{Y}_2 \end{pmatrix} \sim N_2 \left\{ \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2/n_1 & 0 \\ 0 & \sigma_2^2/n_2 \end{pmatrix} \right\},$$

and an application of (3.22) with $a = 0$ and $B^T = (1, -1)$ gives that $\bar{Y}_1 - \bar{Y}_2$ has a normal distribution with mean $\mu_1 - \mu_2$ and variance $n_1^{-1}\sigma_1^2 + n_2^{-1}\sigma_2^2$. To simplify matters, let us suppose that the variances σ_1^2 and σ_2^2 both equal σ^2 , in which case

$$\bar{Y}_1 - \bar{Y}_2 \stackrel{D}{=} (\mu_1 - \mu_2) + \sigma(n_1^{-1} + n_2^{-1})^{1/2}Z,$$

where $Z \sim N(0, 1)$, and $(n_1 - 1)S_1^2/\sigma^2$ and $(n_2 - 1)S_2^2/\sigma^2$ are independent chi-squared variables with $n_1 - 1$ and $n_2 - 1$ degrees of freedom respectively, so

$(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2 \sim \sigma^2 \chi_{n_1+n_2-2}^2$. Hence the *pooled estimate* of σ^2 , S^2 , has distribution given by

$$S^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2} \stackrel{D}{=} \sigma^2 W / (n_1 + n_2 - 2),$$

where $W \sim \chi_{n_1+n_2-2}^2$, independently of $\bar{Y}_1 - \bar{Y}_2$. Consequently the quantity

$$\frac{\bar{Y}_1 - \bar{Y}_2 - (\mu_1 - \mu_2)}{\{S^2(n_1^{-1} + n_2^{-1})\}^{1/2}} \stackrel{D}{=} \frac{Z}{\{W/(n_1 + n_2 - 2)\}^{1/2}} \sim t_{n_1+n_2-2}$$

is a pivot from which confidence intervals for $\mu_1 - \mu_2$ may be determined. The argument parallels that leading to (3.17) and shows that the *two-sample t confidence interval* whose endpoints are

$$(\bar{Y}_1 - \bar{Y}_2) \pm \{S^2(n_1^{-1} + n_2^{-1})\}^{1/2} t_{n_1+n_2-2}(\alpha) \quad (3.24)$$

is a $(1 - 2\alpha)$ confidence interval for $\mu_1 - \mu_2$ based on the two samples. In practice, the random variables in (3.24) are replaced by their observed values, and the resulting interval is given the repeated sampling interpretation.

Example 3.16 (Maize data) For the data in Example 1.1, we have $n_1 = n_2 = 15$, $\bar{y}_1 = 161.5$, $s_1^2 = 837.3$, $\bar{y}_2 = 140.6$ and $s_2^2 = 269.4$. The difference of averages is 20.9 and the pooled estimate of variance is 553.3; note that pooling here ignores the evidence of Example 3.13 that the self-fertilized plants are less variable, that is, $\sigma_2^2 < \sigma_1^2$.

The 0.025 quantile of t_{28} is -2.05 , so the two-sample 0.95 confidence interval for $\mu_1 - \mu_2$ is $20.9 \pm 553.3^{1/2}(1/15 + 1/15)^{1/2} \times 2.05 = (3.34, 38.53)$ eighths of an inch. This confidence interval is slightly narrower than that given in Example 3.11, based on differences of pairs of plants, and gives correspondingly stronger evidence for a height difference in mean heights. However, this interval is less appropriate, both because of the pairing of plants in the original experiment, and because of the evidence for a difference in variances. ■

If there are two normal samples with unequal variances, $\sigma_1^2 \neq \sigma_2^2$, there is no exact pivot. One fairly accurate approach to confidence intervals for the difference of sample means, $\mu_1 - \mu_2$, is based on the approximate pivot

$$T = \frac{\bar{Y}_1 - \bar{Y}_2 - (\mu_1 - \mu_2)}{(S_1^2/n_1 + S_2^2/n_2)^{1/2}} \sim t_v, \quad v = \frac{(S_1^2/n_1 + S_2^2/n_2)^2}{S_1^4/\{n_1^2(n_1 - 1)\} + S_2^4/\{n_2^2(n_2 - 1)\}}.$$

The idea of this is to replace the exact variance of $\bar{Y}_1 - \bar{Y}_2$, $\sigma_1^2/n_1 + \sigma_2^2/n_2$, by an estimate, and then to find the t distribution whose degrees of freedom give the best match to the moments of T .

Example 3.17 (Maize data) For the data in Example 1.1, we have $v = 22.16$, and $t_v(0.025) = -2.07$. Now $s_1^2/n_1 + s_2^2/n_2 = 73.78$, so an approximate 95% confidence interval is $20.9 \pm 2.07 \times 73.78^{1/2}$, that is, $(3.13, 38.74)$. As mentioned before, this interval is more appropriate for these data, but it differs only slightly from the interval in Example 3.16. ■

Joint distribution of $\bar{Y}$ and S^2

We now derive the key result (3.15). The most direct route starts from noting that if $Y_1, \dots, Y_n$ is a random sample from the $N(\mu, \sigma^2)$ distribution, the distribution of $Y = (Y_1, \dots, Y_n)^T$ is $N_n(\mu 1_n, \sigma^2 I_n)$. We now consider the random variable $U = B^T Y$, where the $n \times n$ matrix B^T equals

$$\begin{pmatrix} \frac{1}{n^{1/2}} & \frac{1}{n^{1/2}} & \frac{1}{n^{1/2}} & \frac{1}{n^{1/2}} & \cdots & \frac{1}{n^{1/2}} \\ \frac{1}{2^{1/2}} & -\frac{1}{2^{1/2}} & 0 & 0 & \cdots & 0 \\ \frac{1}{6^{1/2}} & \frac{1}{6^{1/2}} & -\frac{2}{6^{1/2}} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{1}{\{n(n-1)\}^{1/2}} & \frac{1}{\{n(n-1)\}^{1/2}} & \frac{1}{\{n(n-1)\}^{1/2}} & \frac{1}{\{n(n-1)\}^{1/2}} & \cdots & -\frac{n-1}{\{n(n-1)\}^{1/2}} \end{pmatrix}.$$

For $j = 2, \dots, n$, the j th row contains $\{j(j-1)\}^{-1/2}$ repeated $j-1$ times, followed by $-(j-1)\{j(j-1)\}^{-1/2}$ once, with any remaining places filled by zeros. Note that $B^T B = I_n$ and $B^T 1_n = (n^{1/2}, 0, \dots, 0)^T$, which imply that

$$U \sim N_n\{(n^{1/2}\mu, 0, \dots, 0)^T, \sigma^2 I_n\}.$$

Thus the components of U are independent, and only the first, U_1 , has non-zero mean; in fact $U_1 = n^{-1/2} \sum Y_j = n^{1/2} \bar{Y}$, from which we see that $\bar{Y} \sim N(\mu, n^{-1}\sigma^2)$, thus establishing the first line of (3.15). Now

$$\sum_{j=1}^n Y_j^2 = Y^T Y = Y^T B^T B Y = U^T U = \sum_{j=1}^n U_j^2 = n\bar{Y}^2 + U_2^2 + \cdots + U_n^2,$$

which implies that

$$(n-1)S^2 = \sum_{j=1}^n (Y_j - \bar{Y})^2 = \sum_{j=1}^n Y_j^2 - n\bar{Y}^2 = U_2^2 + \cdots + U_n^2.$$

Thus $(n-1)S^2/\sigma^2$ equals the sum of the squares of the $n-1$ standard normal variables $U_2/\sigma, \dots, U_n/\sigma$, and therefore has the chi-squared distribution with $n-1$ degrees of freedom, independent of U_1 , and hence independent of $\bar{Y}$. This establishes the remainder of (3.15).

Exercises 3.2

- 1 Show that the first two derivatives of $\phi(z)$ are $-z\phi(z)$ and $(z^2-1)\phi(z)$. Hence use integration by parts to find the mean and variance of (3.6).
- 2 If X has density (2.7), show that $2\lambda X$ has density (3.9) with $\nu = 2\kappa$.
- 3 Let $h(Z, W)$ be a function of two random variables Z and W whose variance is finite, and let $g(W) = E_W\{h(Z, W) \mid W\}$. Show that $h(Z, W) - g(W)$ has mean zero and is uncorrelated with $g(W)$. Hence establish (3.13).
- 4 Let N be a random variable taking values $0, 1, \dots$, let $G(u)$ be the probability-generating function of N , and let $X_1, X_2, \dots$ be independent variables each having moment-generating function $M(t)$. Use (3.12) to show that $Y = X_1 + \cdots + X_N$ has moment-generating function $G\{M(t)\}$, and hence find the mean and variance of Y in terms of those of X and N .
Use (3.12) and (3.13) to find $E(Y)$ and $\text{var}(Y)$ directly.

- 5 Use (3.6) and (3.9) to derive (3.11).
- 6 Use (3.9) to derive (3.14).
- 7 Check carefully the derivations of (3.8) and (3.10).
- 8 Assuming that the times for each day in Table 2.1 are a random sample from the normal distribution, use the day 2 data to compute (i) a two-sided 0.95 confidence interval for the population mean time in delivery suite and (ii) a 0.95 confidence interval for the population variance. Also give two-sided 0.95 confidence intervals for the difference in mean times for day 1 and day 2, assuming that their variances are (iii) equal and (iv) unequal. Give a 0.95 confidence interval for the ratio of their variances. Repeat (i) and (ii) giving 0.95 upper and lower confidence intervals.
- 9 If $Z \sim N(0, 1)$, derive the density of $Y = Z^2$. Although Y is determined by Z , show they are uncorrelated.
- 10 If $W \sim \chi_v^2$, show that $E(W) = v$, $\text{var}(W) = 2v$ and $(W - v)/\sqrt{2v} \xrightarrow{D} N(0, 1)$ as $v \rightarrow \infty$.
- 11 (a) If $F \sim F_{v_1, v_2}$, show that $1/F \sim F_{v_2, v_1}$. Give the quantiles of $1/F$ in terms of those of F .
(b) Show that as $v_2 \rightarrow \infty$, $v_1 F$ tends in distribution to a chi-squared variable, and give its degrees of freedom.
(c) If Y_1 and Y_2 are independent variables with density e^{-y} , $y > 0$, show that Y_1/Y_2 has the F distribution, and give its degrees of freedom.
- 12 Let $f(t)$ denote the probability density function of $T \sim t_\nu$.
(a) Use $f(t)$ to check that $E(T) = 0$, $\text{var}(T) = \nu/(\nu - 2)$, provided $\nu > 1, 2$ respectively.
(b) By considering $\log f(t)$, show that as $\nu \rightarrow \infty$, $f(t) \rightarrow \phi(t)$.
- 13 If Y and Z are $p \times 1$ and $q \times 1$ vectors of random variables, show that $\text{cov}(Y, Z) = E(YZ^T) - E(Y)E(Z)^T$.
- 14 Verify that if there is a non-zero vector a such that $\text{var}(a^T Y) = 0$, either some Y_r takes a single value with probability one or $Y_r = \sum_{s \neq r} b_s Y_s$, for some r , b_s not all equal to zero.
- 15 Suppose $Y \sim N_p(\mu, \Omega)$ and a and b are $p \times 1$ vectors of constants. Find the distribution of $X_1 = a^T Y$ conditional on $X_2 = b^T Y = x_2$. Under what circumstances does this not depend on x_2 ?
- 16 Otherwise, or by noting that

$$\sigma^{-1} \int \Phi(a + by) \phi\left(\frac{y - \mu}{\sigma}\right) dy = E_Y \{\Pr(Z \leq a + bY \mid Y = y)\},$$

where $Z \sim N(0, 1)$, independent of $Y \sim N(\mu, \sigma^2)$, show that

$$\sigma^{-1} \int \Phi(a + by) \phi\left(\frac{y - \mu}{\sigma}\right) dy = \Phi\left\{\frac{a + b\mu}{(1 + b^2\sigma^2)^{1/2}}\right\}. \quad (3.25)$$

- 17 Let $Y = X_1 + bX_2$, where the X_j are independent normal variables with means μ_j and variances σ_j^2 . Show that conditional on $X_2 = x$, the distribution of Y is normal with mean $\mu_1 + bx$ and variance σ_1^2 , and hence establish that

$$\int \frac{1}{\sigma_1} \phi\left(\frac{y - \mu_1 - bx}{\sigma_1}\right) \frac{1}{\sigma_2} \phi\left(\frac{x - \mu_2}{\sigma_2}\right) dx = \frac{1}{(\sigma_1^2 + b^2\sigma_2^2)^{1/2}} \phi\left\{\frac{y - \mu_1 - b\mu_2}{(\sigma_1^2 + b^2\sigma_2^2)^{1/2}}\right\}.$$

- 18 To establish (3.21), show that the variables $X = Y_1 - \Omega_{12}\Omega_{22}^{-1}Y_2$ and Y_2 have a joint multivariate normal distribution and are independent, find the mean of X , and show that its variance matrix is $\Omega_{11} - \Omega_{12}\Omega_{22}^{-1}\Omega_{21}$. Then use the fact that if X and Y_2 are independent, conditioning on $Y_2 = y_2$ will not change the distribution of X , to give (3.21).

Recall Stirling's formula.

- 19 Let Y have the p -variate multivariate normal distribution with mean vector μ and covariance matrix Ω . Partition Y^T as (Y_1^T, Y_2^T) , where Y_1 has dimension $q \times 1$ and Y_2 has dimension $r \times 1$, and partition μ and Ω conformably. Find the conditional distribution of Y_1 given that $Y_2 = y_2$ direct from the probability density functions of Y and Y_2 .
- 20 Conditional on $M = m$, $Y_1, \dots, Y_n$ is a random sample from the $N(m, \sigma^2)$ distribution. Find the unconditional joint distribution of $Y_1, \dots, Y_n$ when M has the $N(\mu, \tau^2)$ distribution. Use induction to show that the covariance matrix Ω has determinant $\sigma^{2n-2}(\sigma^2 + n\tau^2)$, and show that Ω^{-1} has diagonal elements $\{\sigma^2 + (n-1)\tau^2\}/\{\sigma^2(\sigma^2 + n\tau^2)\}$ and off-diagonal elements $-\tau^2/\{\sigma^2(\sigma^2 + n\tau^2)\}$.

3.3 Simulation

3.3.1 Pseudo-random numbers

Simulation, or the computer generation of artificial data, has many purposes. Among them are:

- to see how much variability to expect in sampling from a particular model. For example, a probability plot for a small sample can be hard to interpret, and in assessing whether any pattern in it is imagined or real it is helpful to compare it with those for sets of simulated data;
- to assess the adequacy of a theoretical approximation. This is illustrated by Figure 2.4, which compares histograms of the average of n simulated exponential variables with the normal density arising from the central limit theorem. The simulations suggest that the approximation is poor when $n \leq 5$, but much improved when $n \geq 20$;
- to check the sensitivity of conclusions to assumptions — for example, how badly do the methods of the previous section fail when the data are not normal? We discuss this in Example 3.24 below;
- to give insight or confirm a hunch, on the principle that a rough answer to the right question is worth more than a precise answer to the wrong question; and
- to provide numerical solutions when analytical ones are unavailable.

The starting point is an algorithm that provides a stream of *pseudo-random* variables, $U_1, U_2, \dots$, supposed independent and uniformly distributed on the interval $(0, 1)$. These are called *pseudo-random* because although the algorithm should ensure that they seem independent and identically distributed, they are predictable to anyone knowing the algorithm. One important class is the *linear congruential generators* defined by

$$X_{j+1} = (aX_j + c) \bmod M, \quad U_j = X_j/M,$$

for some natural number M , with $a, c \in \{0, 1, \dots, M-1\}$; such a generator will repeat with period at most M . The values of M , a and c are chosen to maximize the period and speed of the generator, and the apparent randomness of the output. An example is $M = 2^{48}$, $a = 5^{17}$ and $c = 1$, giving $M/4$ elements of the set $\{0, \dots, M-1\}/M$ in what appears to be a random order.

Some authors call them *quasi-random*.

Not only is it important that the U_j are uniform, but also that they seem independent. One way to do this is to consider k -tuples $(U_j, U_{j+1}, \dots, U_{j+k-1})$ of successive values as points in the set $(0, 1)^k$, where they should be uniformly distributed; see Practical 3.5. Many of the algorithms in standard packages have been thoroughly tested, but it is wise to store the *seed* X_0 so that if necessary the sequence can be repeated, and to perform important calculations using two different generators. Below we suppose it safe to assume that $U_1, U_2, \dots$ are independent identically distributed variables from the $U(0, 1)$ distribution (2.22) and refer to them as random rather than pseudo-random.

Inversion

The simplest way to convert uniform variables into those from other distributions is *inversion*. Let F be the distribution function of a random variable, Y , and let $F^{-1}(u) = \inf\{y : F(y) \geq u\}$. If U has the $U(0, 1)$ distribution (2.22), we saw on page 39 that $Y \stackrel{D}{=} F^{-1}(U)$, and that $F^{-1}(U_1), \dots, F^{-1}(U_n)$ is a random sample from F .

Example 3.18 (Exponential distribution) The distribution function of an exponential random variable with parameter $\lambda > 0$ is

$$F(y) = \begin{cases} 0, & y \leq 0, \\ 1 - \exp(-\lambda y), & 0 < y, \end{cases}$$

and for $0 < u < 1$ the solution to $F(y) = u$ is $y = -\lambda^{-1} \log(1 - u)$. Therefore a random variable from F is $Y = -\lambda^{-1} \log(1 - U) \stackrel{D}{=} -\lambda^{-1} \log U$, because U and $1 - U$ have the same distribution. ■

Example 3.19 (Normal, chi-squared and t distributions) A normal random variable with mean μ and variance σ^2 has distribution function $F(y) = \Phi\{(y - \mu)/\sigma\}$, and therefore $\mu + \sigma \Phi^{-1}(U_1), \dots, \mu + \sigma \Phi^{-1}(U_n)$ is a normal random sample.

If $Z_1, Z_2, \dots$ is a stream of standard normal variables, $V = \sum_{j=1}^v Z_j^2$ is chi-squared with v degrees of freedom, and $T = Z_{v+1}/(V/v)^{1/2}$ has the Student t distribution with v degrees of freedom. Since $Z_j = \Phi^{-1}(U_j)$, V and T are easily obtained. ■

Pseudo-random variables from other distributions and processes can be constructed using their definitions, though statistical packages usually contain specially-programmed algorithms. One general approach for discrete variables is the *look-up method*. Suppose that Y takes values in $\{1, 2, \dots\}$ and that we have created a table containing the values of $\Pi_r = \Pr(Y \leq r)$ and $\pi_r = \Pr(Y = r)$. Then inversion amounts to this algorithm:

- 1 generate $U \sim U(0, 1)$ and set $r = 1$; then
- 2 while $\Pi_r \leq U$ set $r = r + 1$; and finally
- 3 return $Y = r$.

The number of comparisons at step 2 can be reduced by sorting the π_r into decreasing order and re-ordering $\{1, 2, \dots\}$ accordingly. An alternative is to begin searching at

a place that depends on U . Each involves initial expense in obtaining and manipulating the π_r 's, and as the trade-off between this and the number of comparisons is complicated, fast algorithms for discrete distributions can be complex.

Rejection

Inversion is simple, but to be efficient it requires a fast algorithm for F^{-1} . Another approach is *rejection*. Suppose we wish to generate from an awkward density f , and can easily generate from the uniform distribution and from a density g for which $\sup_y f(y)/g(y) = b < \infty$; note that $b > 1$. The rejection algorithm to generate Y from f is:

Sometimes called the acceptance-rejection or envelope method.

- 1 generate X from g and U from the $U(0, 1)$ density, independently;
- 2 set $Y = X$ if $Ubg(X) \leq f(X)$, and otherwise go to 1; finally
- 3 return Y .

To see why this works, note that the interpretation of $\Pr(X \leq a)$ as the area under g to the left of a implies that $(X, Ubg(X))$ is uniformly distributed on the set $\{(x, w) : 0 \leq w \leq bg(x)\}$, and a value Y is returned only if $Ubg(X) \leq f(X)$. For a single pair (X, U) , the probability a value Y is returned and is less than y is

$$\begin{aligned} \Pr\{Ubg(X) \leq f(X) \text{ and } X \leq y\} &= \int_{-\infty}^y \Pr\left\{U \leq \frac{f(X)}{bg(X)} \mid X = x\right\} g(x) dx, \\ &= \int_{-\infty}^y \frac{f(x)}{bg(x)} g(x) dx \\ &= b^{-1} \int_{-\infty}^y f(x) dx, \end{aligned}$$

because U is uniform, independent of X . Hence

$$\begin{aligned} \Pr(Y \leq y \mid \text{value returned}) &= \frac{\Pr\{Ubg(X) \leq f(X) \text{ and } X \leq y\}}{\Pr\{Ubg(X) \leq f(X) \text{ and } X \leq \infty\}} \\ &= \int_{-\infty}^y f(x) dx; \end{aligned}$$

the density of Y is indeed f . The probability a value is returned is b^{-1} , so the algorithm is most efficient when b is as small as possible, and the envelope function $bg(x)$ should ensure both this and fast simulation from g .

Example 3.20 (Half-normal density) A half-normal variable is defined by $Y = |Z|$, where $Z \sim N(0, 1)$. Its density, $f(y) = 2\phi(y)$ for $y > 0$, is shown by the solid line in the left panel of Figure 3.5. The exponential density $g(y) = \lambda e^{-\lambda y}$, declines more slowly than $f(y)$ for large y , and the ratio

$$\frac{f(y)}{g(y)} = \frac{2(2\pi)^{-1/2} e^{-y^2/2}}{\lambda e^{-\lambda y}} = \exp\left\{\lambda y - \frac{1}{2}y^2 + \frac{1}{2} \log\left(\frac{2}{\pi\lambda^2}\right)\right\}$$

is maximized at $y = \lambda$, giving $b = \sup_y f(y)/g(y) = (2/\pi\lambda^2)^{-1/2} e^{\lambda^2/2}$. The function $bg(x)$ with $\lambda = 1$ is shown by the dotted line in the figure.

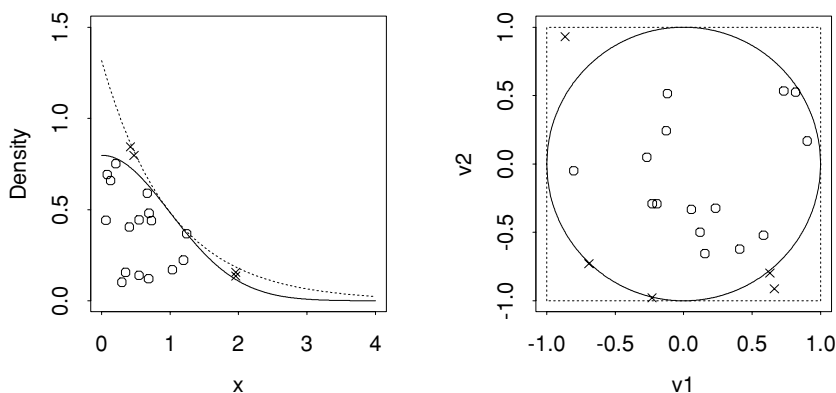


Figure 3.5 Simulation by rejection algorithms. Left panel: half-normal density f (solid) and envelope function bg (dots), with points for which X rejected shown by crosses and those accepted by circles. Right panel: pairs (V_1, V_2) are generated uniformly in the square $[-1, 1] \times [-1, 1]$, but only those in the disk $v_1^2 + v_2^2 \leq 1$ are accepted. They are then transformed into two independent normal variables.

Circles shows pairs $(X, Ubg(X))$ accepted, giving $Y = X$, and crosses show pairs for which X is rejected. These lie in the set $\{(x, w) : f(x) \leq w \leq bg(x)\}$, whose area is $b - 1$, while the area under $bg(x)$ is of course b . The proportion of rejections is minimized by choosing λ to minimize b , and this occurs when $\lambda = 1$, giving $b^{-1} = 0.760$. Whether the resulting algorithm is faster than simply taking $Y = |\Phi^{-1}(U)|$ will depend on the speeds of the functions and the arithmetical operations involved. ■

Rejection can be combined with other methods to give efficient algorithms.

Example 3.21 (Normal distribution) Let Z_1 and Z_2 be two independent standard normal variables. Their joint density is

$$f(z_1, z_2) = \phi(z_1)\phi(z_2) = \frac{1}{2\pi} \exp \left\{ -\frac{1}{2}(z_1^2 + z_2^2) \right\}, \quad -\infty < z_1, z_2 < \infty.$$

The polar coordinates of the point (z_1, z_2) in the plane are $r = (z_1^2 + z_2^2)^{1/2}$ and $\theta = \tan^{-1}(z_2/z_1)$, in terms of which $z_1 = r \cos \theta$, $z_2 = r \sin \theta$. The transformation from (z_1, z_2) to (r, θ) has Jacobian

$$\left| \frac{\partial(z_1, z_2)}{\partial(r, \theta)} \right| = \begin{vmatrix} \cos \theta & \sin \theta \\ -r \sin \theta & r \cos \theta \end{vmatrix} = r > 0,$$

so the joint density of $R = (Z_1^2 + Z_2^2)^{1/2}$ and $\Theta = \tan^{-1}(Z_2/Z_1)$ is

$$f(r, \theta) = f(z_1, z_2) \left| \frac{\partial(z_1, z_2)}{\partial(r, \theta)} \right| = \frac{1}{2\pi} r \exp \left(-\frac{1}{2}r^2 \right), \quad r > 0, 0 \leq \theta < 2\pi.$$

Evidently R and Θ are independent, with Θ uniform on the interval $[0, 2\pi)$ and R having distribution $\Pr(R \leq r) = 1 - \exp(-r^2/2)$. Thus if $U_1, U_2 \stackrel{\text{iid}}{\sim} U(0, 1)$, we can generate Z_1 and Z_2 by setting $Z_1 = R \cos \Theta$, $Z_2 = R \sin \Theta$, where $\Theta = 2\pi U_1$ and $R = (-2 \log U_2)^{1/2}$; this amounts to inversion for R and Θ .

A drawback of this method is that trigonometric functions such as $\sin(\cdot)$ and $\cos(\cdot)$ tend to be slow. It is better to avoid them by using rejection, as follows. We first generate $U_1, U_2 \stackrel{\text{iid}}{\sim} U(0, 1)$ and set $V_1 = 2U_1 - 1$ and $V_2 = 2U_2 - 1$; (V_1, V_2) is uniformly

distributed in the square $[-1, 1] \times [-1, 1]$. If $S = V_1^2 + V_2^2 > 1$, we reject (V_1, V_2) and start again; see the right panel of Figure 3.5. If it is accepted, the point (V_1, V_2) is uniform in the unit disk, S is independent of the angle $\Theta = \tan^{-1}(V_2/V_1)$ by symmetry, and comparison of areas gives $\Pr(S \leq s) = (s\pi)/\pi = s$, $0 \leq s \leq 1$, so $S \sim U(0, 1)$; this implies that $R \stackrel{D}{=} (-2 \log S)^{1/2}$. Furthermore, if (V_1, V_2) has been accepted, then $\cos \Theta = V_1/S^{1/2}$, $\sin \Theta = V_2/S^{1/2}$. Then $Z_1 = R \cos \Theta = V_1(-2S^{-1} \log S)^{1/2}$ and $Z_2 = R \sin \Theta = V_2(-2S^{-1} \log S)^{1/2}$ are independent standard normal variables, and may be obtained without recourse to trigonometric functions. The efficiency of this algorithm is $\pi/4 \doteq 0.785$.

If $Z_1, Z_2 \stackrel{\text{iid}}{\sim} N(0, 1)$, then their ratio $C = Z_2/Z_1$ has a Cauchy distribution. Thus if we want to generate a Cauchy variable, we need only take $R \sin \Theta / (R \cos \Theta) = V_2/V_1$, where (V_1, V_2) lies inside the unit disk. This suggests the *ratio of uniforms* method (Problem 3.7). ■

It may be hard to find an envelope density $g(y)$ for $f(y)$, leading to a high initialization cost for rejection sampling. If $f(y)$ is log-concave, however, so $h(y) = \log f(y)$ is concave in y , then it turns out to be easy to find an envelope from which quick simulation is possible. To see how, let $f(y)$ be a log-concave density with known support $[y_L, y_U]$, where possibly $y_L = -\infty$ or $y_U = \infty$ or both. Then for any y_1, y_2 in $[y_L, y_U]$,

$$h\{\gamma y_1 + (1 - \gamma)y_2\} \geq \gamma h(y_1) + (1 - \gamma)h(y_2), \quad 0 \leq \gamma \leq 1,$$

and if $h(y)$ is piecewise differentiable, as we henceforth assume, then $h'(y) = dh(y)/dy$ is monotonic decreasing in y , though perhaps $h(y)$ has straight line segments or $h'(y)$ is discontinuous.

Let $y_L \leq y_1 < \dots < y_k \leq y_U$ and suppose that $h(y_1), \dots, h(y_k)$ and $h'(y_1), \dots, h'(y_k)$ are known. If $y_L = -\infty$ we choose y_1 so that $h'(y_1) > 0$. Likewise if $y_U = \infty$, we choose y_k so that $h'(y_k) < 0$. We then define a function $h_+(y)$ by taking the upper boundary of the convex hull generated by the tangents to $h(y)$ at $y_1, \dots, y_k$; see Figure 3.6. That is,

$$h_+(y) = \begin{cases} h(y_1) + (y - y_1)h'(y_1), & y_L < y \leq z_1, \\ h(y_{j+1}) + (y - y_{j+1})h'(y_{j+1}), & z_j \leq y \leq z_{j+1}, \quad j = 1, \dots, k-1, \\ h(y_k) + (y - y_k)h'(y_k), & z_k \leq y < y_U, \end{cases}$$

where

$$z_j = y_j + \frac{h(y_j) - h(y_{j+1}) + (y_{j+1} - y_j)h'(y_{j+1})}{h'(y_{j+1}) - h'(y_j)}, \quad j = 1, \dots, k-1,$$

are the values of y at which the tangents at y_j and y_{j+1} intersect; we also set $z_0 = y_L$ and $z_k = y_U$. As the density $g_+(y) \propto \exp\{h_+(y)\}$ consists of k piecewise exponential portions, a variable X with density g_+ may be generated by inversion and then rejection applied. If the X thus generated is rejected, then $h(X)$ and $h'(X)$ can be used to update h_+ and provide a better envelope for subsequent simulation.

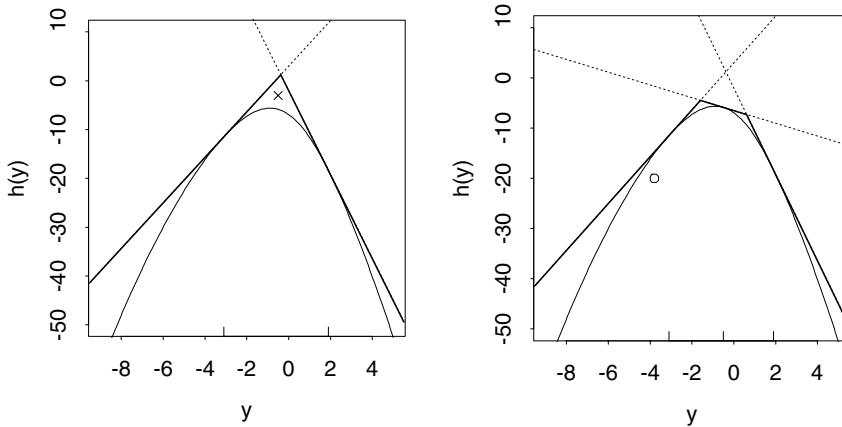


Figure 3.6 Adaptive rejection sampling from log-concave density proportional to $h(y)$ (solid). The left panel shows the initial envelope (heavy), formed as the concave hull of tangents (dotted) to $h(y)$ at $y = -3.1, 1.9$ (rug). The envelope density looks like two exponential densities, back to back, from which a value shown by a cross is generated. This value is rejected but used to update the envelope to that on the right, so the corresponding density has three exponential parts. This time the value generated by rejection sampling (circle) is accepted.

This discussion suggests an *adaptive rejection sampling* algorithm:

1. Initialize by choosing $y_1 < \dots < y_k$, calculating $h(y_1), \dots, h(y_k)$ and $h'(y_1), \dots, h'(y_k)$, $h_+(y)$ and $g_+(y)$. Then
2. generate independent variables X from g_+ and U from the $U(0, 1)$ density. If $U \leq \exp\{h(X) - h_+(X)\}$ then set $Y = X$ and return Y ; otherwise
3. replace k by $k + 1$, update $y_1, \dots, y_k$, $h(y_1), \dots, h(y_k)$ and $h'(y_1), \dots, h'(y_k)$ by adding X , $h(X)$ and $h'(X)$, recompute $h_+(y)$ and $g_+(y)$ and go to 2.

This can be accelerated by using $h(y_1), \dots, h(y_k)$ and $h'(y_1), \dots, h'(y_k)$ to add a lower envelope $h_-(y)$ and then accepting X if $U \leq \exp\{h_-(X) - h_+(X)\}$, in which case $h(X)$ need not be computed (Problem 3.12).

Example 3.22 (Adaptive rejection) To illustrate this we take

$$h(y) = ry - m \log(1 + e^y) - \frac{(y - \mu)^2}{2\sigma^2} + c, \quad -\infty < y < \infty,$$

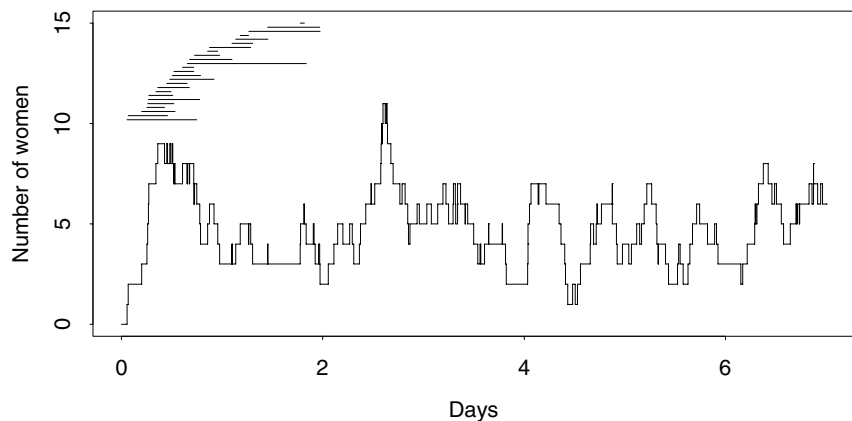
where $m, \sigma^2 > 0$ and c is the constant ensuring that $\exp\{h(y)\}$ has unit integral; see Example 11.26. As we deal only with ratios of densities we can ignore c below, and Figure 3.6 shows $h(y)$ for $r = 2$, $m = 10$, $\mu = 0$, $\sigma^2 = 1$ and when we set $c = 0$; here $y_L = -\infty$ and $y_U = \infty$.

An initial search establishes that $h'(-3.1) > 0$ and $h'(1.9) < 0$, and the resulting envelope is shown in the left panel. The corresponding density $g_+(y)$ looks like two back-to-back exponential densities, from which it is easy to simulate a value X . This is accepted if $Ug_+(X) < h(X)$, where $U \sim U(0, 1)$. In the event, the value -0.5 is generated but not accepted, and the envelope is updated to that shown in the right panel. A value generated from the new $g_+(y)$ is accepted, terminating the algorithm. Otherwise the envelope would again be updated, and the process repeated. ■

Applications

Fast, tested generators are available in many statistical packages, so the details can often — but not always — be ignored. Here are two uses of them.

Figure 3.7 Numbers of women in the delivery suite over a week of simulations from the model for the birth data. Also shown are arrival and departure times for the first 25 simulated women.



Example 3.23 (Birth data) The data in Example 2.3 were collected in order to assess the workload in the delivery suite. Examples 2.11 and 2.12 suggest that the daily number of arrivals leading to normal deliveries is Poisson with mean about $\lambda = 12.9$, and that each woman remains for a period whose density is roughly gamma with shape $\alpha = 3.15$ and mean $\mu = 7.93$ hours, independent of the others. To simulate t days of data from this model, we generate a Poisson random variable N with mean λ for each day, and then generate N arrival times uniformly through the day. We create departure times by adding a gamma variable with mean μ and shape α to each arrival time; of course a woman may not depart on the day she arrived. We repeat this for each day, and record how many women are present at each arrival and departure.

Figure 3.7 shows a week of simulated workload. Note the initial ‘burn-in’ period, due to starting with no women present rather than in steady state. The number present has long-run average $12.9 \times 7.93/24 = 4.26$, but it fluctuates widely, with bursts of activity when several women arrive almost together.

Such simulations show the random variation in the process due to the model, but they do not reflect the fact that the model itself is uncertain, because it has been estimated. However it would be easy to change λ , α , and μ , or to replace the gamma by a different distribution, and then to repeat the simulation. This would help assess the effect of model uncertainty.

On leaving the delivery suite, women and their babies go to a ward where midwives give post-natal care. At one stage hospital managers hoped to save money by imposing a rigid demarcation between ward and delivery suite, but this would have been counter-productive. According to hospital guidelines, each woman in the delivery suite should have a midwife with her at all times, so when bursts of activity begin it is essential to be able to call in midwives immediately. It is more expensive to do so from outside, so costs are reduced by allowing easy transfer of workers between ward and suite. ■

The previous example illustrates a particularly simple queueing system — each ‘customer’ must be dealt with at once, so there is no queue! More complicated queues arise in many contexts, and *discrete-event simulation* packages exist to help operations researchers estimate quantities such as the average waiting-time.

We now use simulation use to assess properties of a statistical procedure.

Example 3.24 (*t* statistic) The elements of a random sample $Y_1, \dots, Y_n$ from the $N(\mu, \sigma^2)$ density may be expressed $Y_j = \mu + \sigma Z_j$, where the Z_j are standard normal variables. The t statistic may be written as

Recall that $\bar{Y}$ and S^2 are the average and sample variance of $Y_1, \dots, Y_n$.

$$T = \frac{\bar{Y} - \mu}{(S^2/n)^{1/2}} = \frac{n^{1/2}(\mu + \sigma \bar{Z} - \mu)}{\{(n-1)^{-1}\sigma^2 \sum_j (Z_j - \bar{Z})^2\}^{1/2}} = \frac{n^{1/2}\bar{Z}}{S_Z},$$

say, whether or not the Z_j are normal. When they are, T has a Student t distribution on $n-1$ degrees of freedom and its quantiles $t_{n-1}(\alpha)$ may be explicitly calculated, leading to the exact $(1-2\alpha)$ confidence interval (3.17). How badly does that interval fail when the data are not normal?

Suppose the Z_j have mean zero but distribution F otherwise unspecified. Then the confidence interval (3.17) contains μ with probability

$$\Pr\{\bar{Y} - n^{-1/2}St_{n-1}(1-\alpha) \leq \mu \leq \bar{Y} - n^{-1/2}St_{n-1}(\alpha)\}, \quad (3.26)$$

and this equals

$$\begin{aligned} \Pr\{t_{n-1}(\alpha) \leq T \leq t_{n-1}(1-\alpha)\} &= \Pr\left\{t_{n-1}(\alpha) \leq \frac{n^{1/2}\bar{Z}}{S_Z} \leq t_{n-1}(1-\alpha)\right\} \\ &= p(1-\alpha, n, F) - p(\alpha, n, F), \end{aligned}$$

say, where

$$p(\alpha, n, F) = \Pr\left\{\frac{n^{1/2}\bar{Z}}{S_Z} \leq t_{n-1}(\alpha)\right\}.$$

When F is normal, $p(\alpha, n, F) = \alpha$ and (3.26) is $(1-2\alpha)$, as it should be.

Given any F , α and n , we estimate $p(\alpha, n, F)$ thus. For $r = 1 \dots, R$,

- generate $Z_1, \dots, Z_n \stackrel{\text{iid}}{\sim} F$;
- calculate $T_r = n^{1/2}\bar{Z}/S_Z$; then
- set $I_r = I\{T_r \leq t_{n-1}(\alpha)\}$.

$I\{A\}$ is the indicator of the event A .

Having obtained $I_1, \dots, I_R$, we compute $\hat{p} = R^{-1} \sum_r I_r$, whose expectation is

$$\mathbb{E}\left(R^{-1} \sum_{r=1}^R I_r\right) = \mathbb{E}\left[I\left\{\frac{n^{1/2}\bar{Z}}{S_Z} \leq t_{n-1}(\alpha)\right\} \middle| Z_1, \dots, Z_n \stackrel{\text{iid}}{\sim} F\right] = p(\alpha, n, F).$$

Now $\sum_r I_r$ is binomial with denominator R and probability $p(\alpha, n, F)$, so $\hat{p}$ has variance $R^{-1}p(\alpha, n, F)\{1 - p(\alpha, n, F)\}$. This can be used to gauge the value of R needed to estimate $p(\alpha, n, F)$ with given precision. For example, if $p(\alpha, n, F) \doteq \alpha = 0.05$, then $R = 1600$ gives standard deviation roughly $\{0.05(1-0.05)/1600\}^{1/2} = 0.0054$, and a crude 95% confidence interval for $p(\alpha, n, F)$ is $\hat{p} \pm 0.01$.

Table 3.2 shows values of $100\hat{p}$ for various distributions F , using $n = 10$ and $R = 1600$. The second and third columns are for $\alpha = 0.05, 0.95$, while the fourth shows the estimated probability that the confidence interval contains μ ; ideally this

Table 3.2 Estimated coverage probabilities $p(\alpha, n, F)$, $p(1 - \alpha, n, F)$, and $p(1 - \alpha, n, F) - p(\alpha, n, F)$, for $\alpha = 0.05$ and 0.025 , for 1600 samples of size $n = 10$ from various distributions. The Laplace and mixture densities are $\frac{1}{2} \exp(-|z|)$ and $0.9\phi(z) + 0.1\phi(z/3)/3$, for $z \in \mathbb{R}$, and t_ν denotes the t density on ν degrees of freedom. The ‘slash’ distribution is that of Z/U , where $Z \sim N(0, 1)$ and $U \sim U(0, 1)$ independently. The estimates have been multiplied by 100 for convenience and have standard errors of about 0.5.

<i>F</i>	Target					
	5	95	90	2.5	97.5	95
Normal	4.9	94.7	88.8	2.6	97.1	94.4
Laplace	4.1	94.9	90.8	2.2	98.1	95.9
Mixture	4.0	94.9	90.9	1.9	98.0	96.1
t_{20}	5.4	95.4	90.1	2.2	97.7	95.5
t_{10}	6.1	93.9	87.8	2.6	97.0	94.4
t_5	4.6	95.3	90.7	2.5	98.1	95.6
t_1 (Cauchy)	2.3	97.3	95.1	0.8	99.1	98.3
Slash	2.6	97.4	94.9	1.3	99.3	98.0
Gamma, $\alpha = 2$	9.7	97.9	88.3	6.3	99.1	92.8

would be $1 - 2\alpha = 0.90$. Columns 5–7 give the same quantities for 95% confidence intervals. The first row is included to check the simulation: it does not hit the target exactly, due to simulation randomness, but it is close. Laplace, mixture, ‘slash’ and t_ν densities have heavier tails than the normal; the mixture corresponds to $N(0, 1)$ samples that are occasionally contaminated by $N(0, 3^2)$ variables. The results suggest that heavy-tailed data have little effect on the probabilities until the extreme cases $\nu = 1$ and the ‘slash’ distribution, for both of which the Z_j have infinite mean. Then the intervals are too wide and therefore have too great a chance of containing μ . The gamma distribution is the only asymmetric case, and this shows in the estimated one-tailed probabilities p , though the estimates of $p(1 - \alpha, n, F) - p(\alpha, n, F)$ remain reasonably close to $(1 - 2\alpha)$. Overall the performance of T seems fairly satisfactory unless the data are grossly non-normal.

Simulation timings depend on the computer and language used, as well as the skill of the programmer, so they are often uninformative. Having said this, it took about 20 seconds to obtain each row of the table, using about 25 lines of code in total. This compares very favourably with the time and effort that would be involved in getting such results analytically. ■

3.3.2 Variance reduction

Even though it involves no chemicals or nasty smells, a simulation experiment is nonetheless an experiment, and it may be worth considering how to increase its precision for a given effort. There are numerous ways to do this, but as they all involve extra work on the part of the experimenter, they are only worthwhile when the amount of simulation is large: a reduction from 30 to five seconds matters much less than one from 30 to five days.

Suppose that we wish to estimate properties of a rather awkward statistic $T = t(Y_1, \dots, Y_n)$ that is correlated with a statistic $W = w(Y_1, \dots, Y_n)$ with known properties. Then one way to use W is to write $T = W + (T - W) = W + D$, say, work out the relevant properties of the *control variate* W analytically, and use simulation only for the difference D . For example, if moments of W are available explicitly but

This section may be skipped on a first reading.

<i>F</i>		<i>p</i>					
		0 (Average)	0.1	0.2	0.3	0.4	0.5 (Median)
Normal	<i>nvar</i> (<i>T</i>)	1	1.05	1.13	1.23	1.35	1.54
	Correlation	1	0.98	0.95	0.91	0.86	0.81
	Efficiency gain	∞	10.4	4.9	3.1	2.3	1.9
<i>t</i> ₅	<i>nvar</i> (<i>T</i>)	1.67	1.38	1.37	1.42	1.53	1.73
	Correlation	1	0.93	0.89	0.84	0.80	0.75
	Efficiency gain	∞	2.1	1.4	1.1	1	0.9

Table 3.3 Estimated variances of $p \times 100\%$ trimmed averages in samples of size $n = 21$ from the normal and t_5 distributions.

we want to estimate the variance of T , we write

$$\text{var}(T) = \text{var}(W) + 2\text{cov}(W, D) + \text{var}(D),$$

where only terms involving D need to be estimated by simulation. We then generate R independent samples $Y_1, \dots, Y_n$ and calculate T , W and D for each, giving (T_r, W_r, D_r) , $r = 1, \dots, R$. Then $\text{var}(T)$ is estimated by

$$V_1 = \text{var}(W) + \frac{2}{R-1} \sum_{r=1}^R (W_r - \overline{W})(D_r - \overline{D}) + \frac{1}{R-1} \sum_{r=1}^R (D_r - \overline{D})^2, \tag{3.27}$$

where the exact quantity $\text{var}(W)$ replaces the sample variance of $W_1, \dots, W_R$. The usual estimate of $\text{var}(T)$ would be $V_2 = (R-1)^{-1} \sum_r (T_r - \overline{T})^2$. If $\text{var}(W)$ is a large part of $\text{var}(T)$, then $\text{var}(V_1)$ may be much smaller than $\text{var}(V_2)$, but the *efficiency gain* $\text{var}(V_2)/\text{var}(V_1)$ will depend on the correlation between W and T .

Example 3.25 (Trimmed average) Let $Y_1, \dots, Y_n$ be a random sample from a distribution F with mean μ and variance σ^2 . One estimate of μ is the sample average $\overline{Y}$, but as this is sensitive to bad values it may be preferable to use the $p \times 100\%$ trimmed average

$$T = (n - 2k)^{-1} \sum_{j=k+1}^{n-k} Y_{(j)},$$

where $Y_{(1)} \leq \dots \leq Y_{(n)}$ are the order statistics of the sample and $k = pn$ is an integer. One measure of the precision of T is its variance, and if we found that $\text{var}(T) < \text{var}(\overline{Y})$ for many different distributions F , we might choose to use T rather than $\overline{Y}$. Given F , $\text{var}(T)$ can in principle be obtained exactly, but as the calculations are tedious it is simpler to simulate.

An obvious control variate is $W = \overline{Y} = n^{-1} \sum_j Y_{(j)}$, which has variance σ^2/n and is perfectly correlated with T if $p = 0$. We simulate as described above, obtaining R values of W_r, T_r and $D_r = T_r - W_r$, and estimate $\text{var}(T)$ using (3.27). Table 3.3 shows values of nV_1 for samples of size $n = 21$ from the normal and the t_5 distribution, using various values of p ; we took $R = 1000$ replicates. The table also shows the estimated correlation between W and T , and the efficiency gains due to use of control variates, estimated by repeating the experiment 50 times. In practice one would have just one

value of V_1 and one of V_2 ; the repetition here was needed only to find the efficiency gains. These are largest when p is small, and even infinite when $p = 0$, when $W = T$ and $\text{var}(V_2) = 0$. In this case $D = T - W = 0$, and as $\text{var}(W) = \text{var}(\bar{Y})$ is known exactly, V_1 is constant and hence $\text{var}(V_1) = 0$; simulation is then unnecessary. The efficiency gains depend not only on the correlation between W and T , but also on the underlying distribution F .

For normal data, the increase in variance when using T rather than $\bar{Y}$ is modest for $p < 0.3$, and for t_5 data $\text{var}(T) < \text{var}(\bar{Y})$ when $0 < p < 0.5$. This suggests that a lightly trimmed average may be preferable to $\bar{Y}$ for non-normal data and not much more variable than $\bar{Y}$ for normal data, but we would need more extensive results to be sure. ■

Importance sampling

Another approach to variance reduction is *importance sampling*. The key idea here is that sometimes most of the sampling is unproductive, and then it is better to concentrate on the parts of the sample space where it is most valuable. The idea is often used in Monte Carlo integration. Suppose we want to estimate

$$\psi = E\{m(Y)\} = \int m(y)g(y)dy.$$

The direct approach is to generate $Y_1, \dots, Y_R$ independently from density g , and to set $\hat{\psi} = R^{-1} \sum_r m(Y_r)$. This has mean and variance

$$E(\hat{\psi}) = E\{m(Y)\} = \psi, \quad \text{var}(\hat{\psi}) = \int m(y)^2 g(y) dy - \psi^2,$$

but it may be a very poor estimate. For example, if $m(Y) = I(Y \leq a)$ and $\psi = \Pr(Y \leq a)$ is very small, then most of the Y_r will not contribute to $\hat{\psi}$, and the effort spent in generating them will be wasted. Instead we try simulating from a density h , chosen to concentrate effort in the important part of the sample space; the support of h must include the support of g . The resulting estimator is the *raw importance sampling estimator* $\hat{\psi}_{\text{raw}} = R^{-1} \sum_r m(Y_r)w(Y_r)$, where $W = w(Y) = g(Y)/h(Y)$ is known as the *importance sampling weight*. The mean and variance of $\hat{\psi}_{\text{raw}}$ are

$$\begin{aligned} E(\hat{\psi}_{\text{raw}}) &= E_h\{m(Y)w(Y)\} = \int m(y) \frac{g(y)}{h(y)} h(y) dy = \int m(y)g(y) dy = \psi, \\ \text{var}(\hat{\psi}_{\text{raw}}) &= R^{-1} \text{var}_h\{m(Y)w(Y)\} \\ &= R^{-1} [E_h\{m(Y)^2 w(Y)^2\} - E_h\{m(Y)w(Y)\}^2] \\ &= R^{-1} \left\{ \int m(y)^2 \frac{g(y)}{h(y)} g(y) dy - \psi^2 \right\}. \end{aligned}$$

Hence $\hat{\psi}_{\text{raw}}$ will be a big improvement on $\hat{\psi}$ if

$$\frac{\text{var}(\hat{\psi})}{\text{var}(\hat{\psi}_{\text{raw}})} = \frac{\int m(y)^2 g(y) dy - \psi^2}{\int m(y)^2 \frac{g(y)}{h(y)} g(y) dy - \psi^2} \quad (3.28)$$

is large. This ratio depends on h , a bad choice of which can make $\hat{\psi}_{\text{raw}}$ much more variable than is $\hat{\psi}$. The trick is to choose h well.

The support of a density f is $\{y : f(y) > 0\}$.

E_h denotes expectation with respect to density h .

z	-3	-2	-1	0	1	2	3
μ_z	-3.15	-2.22	-1.34	-0.62	-0.18	-0.03	-0.002
Efficiency gain	222	19	4.1	1.75	1.19	1.04	1.004

Table 3.4 Efficiency gains in importance sampling to estimate normal probability $\Phi(z)$. μ_z is the optimal tilting parameter.

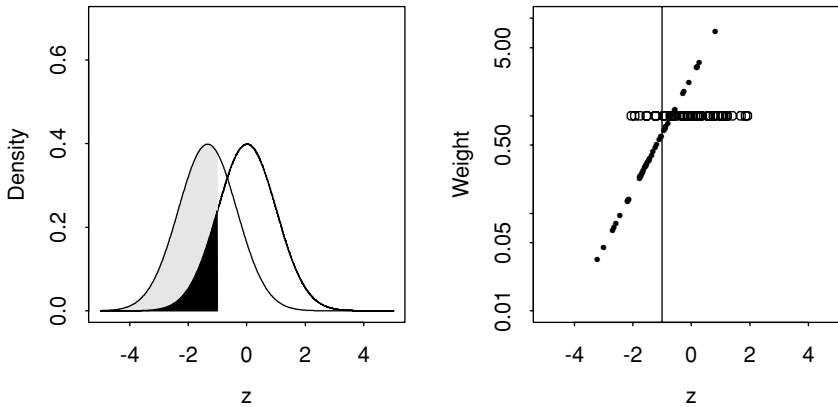


Figure 3.8 Importance sampling for normal tail probability. Left: $N(0, 1)$ density and area $\Phi(z)$ to be estimated (heavy shading), with importance sampling density $N(\mu_z, 1)$, whose lightly shaded area contributes to $\hat{\psi}_{\text{raw}}$. Right: weights for samples with $R = 50$ from $N(0, 1)$ (circles) and from $N(\mu_z, 1)$ (blobs). The vertical line shows $z = -1$; only points to the left of that line contribute to estimation of $\Phi(z)$.

Example 3.26 (Normal probability) Marooned on a desert island with only parrots for company, a shipwrecked statistician decides to realize his lifelong ambition of memorizing values of the normal integral $\Phi(z)$; he hopes to make himself more attractive to the statisticienne of his dreams. His statistical tables have been ruined by salt water, but washed up on the beach he finds a programmable solar-powered calculator on which he is able to implement a slow but reliable normal random number generator.

Rather than estimate $\psi = \Phi(z)$ directly, he decides to use importance sampling from the $N(\mu, 1)$ distribution, taking $m(Y) = I(Y \leq z)$, $g(y) = \phi(y)$, and $h(y) = \phi(y - \mu)$. If $Y_1, \dots, Y_R \stackrel{\text{iid}}{\sim} g$, then $\hat{\psi} = R^{-1} \sum I(Y_r \leq z)$ has mean $\Phi(z)$ and variance $R^{-1} \Phi(z)\{1 - \Phi(z)\}$. If he samples from h , the importance sampling estimate is $\hat{\psi}_{\text{raw}} = R^{-1} \sum_r w(Y_r) I(Y_r \leq z)$, where $w(y) = \phi(y)/\phi(y - \mu) = \exp(\frac{1}{2}\mu^2 - \mu y)$, and it turns out that

$$\text{var}(\hat{\psi}_{\text{raw}}) = R^{-1} \{\exp(\mu^2) \Phi(z + \mu) - \Phi(z)^2\}. \quad (3.29)$$

Given z , therefore, the optimal value μ_z of μ minimizes $e^{\mu^2} \Phi(z + \mu)$. Table 3.4 shows values of μ_z and the efficiency gain (3.28) for a few values of z . Note how $\mu_z \doteq z$ for $z < 0$, but not for $z > 0$, and how importance sampling becomes increasingly effective as $z \rightarrow -\infty$, when almost none of the Y_r contribute to $\hat{\psi}$. For $z > 0$, most of the observations contribute to $\hat{\psi}$ and importance sampling gives little improvement.

The panels of Figure 3.8 show the optimal importance sampling distribution when $z = -1$ and the weights obtained in samples of size $R = 50$ from the $N(0, 1)$ and $N(\mu_z, 1)$ distributions. Most of the observations generated from $\phi(y - \mu_z)$ contribute to $\hat{\psi}_{\text{raw}}$, whereas only a few of those from $\phi(y)$ contribute to $\hat{\psi}$. The efficiency gain

of 4.1 implies that 50 observations from the $N(\mu_z, 1)$ distribution are worth about 200 from the $N(0, 1)$ distribution. The gains are larger when $z \rightarrow -\infty$, and combined with the fact that $\Phi(z) = 1 - \Phi(-z)$ should enable our hero to fulfil his ambition before he is rescued. ■

A difficulty with $\hat{\psi}_{\text{raw}}$ is that the weights W_r can be very variable, with one or two large ones dominating the rest, leading to the average weight $\bar{W}$ being very different from its expectation $E_h(W) = 1$. This can be dealt with by rescaling the weights to $W'_r = W_r/\bar{W}$, for which $\bar{W}' = 1$, resulting in the *importance sampling ratio estimator* $\hat{\psi}_{\text{rat}} = R^{-1} \sum_r W'_r m(Y_r)$. Another approach treats W as a control variate, assuming that the pair $(T, W) = (m(Y)w(Y), w(Y))$ has approximately a bivariate normal distribution, and then estimating the conditional mean of T given $W = 1$. This results in the *importance sampling regression estimator*

$$\hat{\psi}_{\text{reg}} = \bar{T} + \frac{\sum_r (W_r - \bar{W})(T_r - \bar{T})}{\sum_r (W_r - \bar{W})^2} (1 - \bar{W}), \quad T_r = m(Y_r)w(Y_r);$$

note that $\hat{\psi}_{\text{raw}} = \bar{T}$. If T and W are positively correlated, the ratio here will be positive, and if $\bar{W} > 1$ the adjustment reduces $\hat{\psi}_{\text{raw}}$ by an amount that depends on $1 - \bar{W}$. This makes sense because if T and W are positively correlated and $\bar{W} > E(W) = 1$, then it is likely that $\bar{T} > E(T)$. Both ratio and regression estimators tend to improve on $\hat{\psi}_{\text{raw}}$.

Exercises 3.3

- 1 Show how to use inversion to generate Bernoulli random variables. If $0 < \pi < 1$, what distribution has $\sum_{j=1}^m I(U_j \leq \pi)$?
- 2 Write down algorithms to generate values from the gamma density with small integer shape parameter by (a) direct construction using exponential variables, (b) rejection sampling with an exponential envelope.
- 3 The Cholesky decomposition of an $p \times p$ symmetric positive matrix Ω is the unique lower triangular $p \times p$ matrix L such that $LL^T = \Omega$. Find the distribution of $\mu + LZ$, where Z is a vector containing a standard normal random sample $Z_1, \dots, Z_p$, and hence give an algorithm to generate from the multivariate normal distribution.
- 4 If inversion can be used to generate a variable Y with distribution function F , discuss how to generate values from F conditioned on the events (a) $Y \leq y_U$, (b) $y_L < Y \leq y_U$. Under what circumstances might rejection sampling be sensible?
Define Z by setting $Z = j$ when $Y \leq y_j$, for $y_1 < \dots < y_{k-1} < y_k = \infty$. Give an algorithm to generate Z .
- 5 If X has density $\lambda e^{-\lambda x}$, $x > 0$, show that $\Pr(r-1 \leq X \leq r) = e^{-\lambda(r-1)}(1 - e^{-\lambda})$. If Y has geometric density $\Pr(Y = r) = \pi(1 - \pi)^{r-1}$, for $r = 1, 2, \dots$ and $0 < \pi < 1$, show that $Y \stackrel{D}{=} \lceil \log U / \log(1 - \pi) \rceil$. Hence give an algorithm to generate geometric variables.
- 6 Construct a rejection algorithm to simulate from $f(x) = 30x(1-x)^4$, $0 \leq x \leq 1$, using the $U(0, 1)$ density as the proposal function g . Give its efficiency.
- 7 Verify (3.29).

3.4 Bibliographic Notes

The idea of a confidence interval belongs to statistical folklore, but its mathematical formulation and the repeated sampling interpretation were developed by J. Neyman in the 1930s. Fisher argued strongly against the repeated sampling interpretation and developed his own approaches based on conditioning and fiducial inference. Welsh (1996) gives a thoughtful comparison of these and other approaches to inference.

Inference procedures for normal samples are treated in many basic statistics texts.

Stochastic simulation is a very large topic. In addition to books such as Rubinstein (1981), Fishman (1996), Morgan (1984), Ripley (1987), and Robert and Casella (1999), there is a rapidly growing literature on simulation for stochastic processes, often using Markov chain theory; see the bibliographic notes to Chapter 11.

Jerzy Neyman (1894–1981) was born in Moldavia and studied mathematics at Kharkov University and then statistics in Warsaw and University College London, where he worked on the basis of hypothesis testing with Egon Pearson, on experimental design, and on sampling theory. In 1938 he moved to Berkeley and became a leading figure in the development of statistics in the USA.

3.5 Problems

- 1 Suppose that $Y_1, \dots, Y_4$ are independent normal variables, each with variance σ^2 , but with means $\mu + \alpha + \beta + \gamma, \mu + \alpha - \beta - \gamma, \mu - \alpha + \beta - \gamma, \mu - \alpha - \beta + \gamma$. Let

$$Z^T = \frac{1}{4}(Y_1 + Y_2 + Y_3 + Y_4, Y_1 + Y_2 - Y_3 - Y_4, Y_1 - Y_2 + Y_3 - Y_4, Y_1 - Y_2 - Y_3 + Y_4).$$

Calculate the mean vector and covariance matrix of Z , and give the joint distribution of Z_1 and $V = Z_2^2 + Z_3^2 + Z_4^2$ when $\alpha = \beta = \gamma = 0$. What is then the distribution of $Z_1/(V/3)^{1/2}$?

- 2 W_i, X_i, Y_i , and $Z_i, i = 1, 2$, are eight independent, normal random variables with common variance σ^2 and expectations μ_W, μ_X, μ_Y and μ_Z . Find the joint distribution of the random variables

$$T_1 = \frac{1}{2}(W_1 + W_2) - \mu_W, \quad T_2 = \frac{1}{2}(X_1 + X_2) - \mu_X,$$

$$T_3 = \frac{1}{2}(Y_1 + Y_2) - \mu_Y, \quad T_4 = \frac{1}{2}(Z_1 + Z_2) - \mu_Z,$$

$$T_5 = W_1 - W_2, \quad T_6 = X_1 - X_2, \quad T_7 = Y_1 - Y_2, \quad T_8 = Z_1 - Z_2.$$

Hence obtain the distribution of

$$U = 4 \frac{T_1^2 + T_2^2 + T_3^2 + T_4^2}{T_5^2 + T_6^2 + T_7^2 + T_8^2}.$$

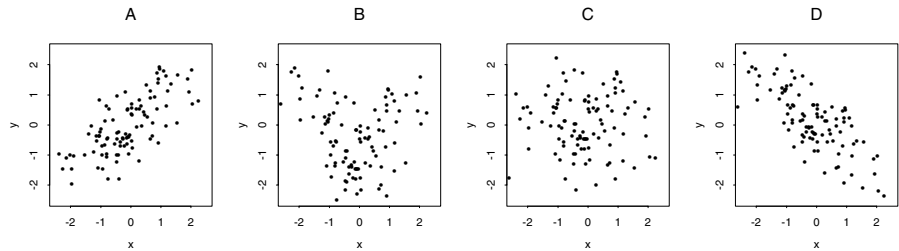
Show that the random variables $U/(1+U)$ and $1/(1+U)$ are identically distributed, without finding their probability density functions. Find their common density function and hence determine $\Pr(U \leq 2)$.

- 3 Figure 3.9 shows samples of size 100 from densities in which (i) X and Y are independent; (ii) $\text{corr}(X, Y) = -0.7$; (iii) $\text{corr}(X, Y) = 0.7$; (iv) $\text{corr}(X, Y) = 0$. Say which is which and why.
- 4 (a) Suppose that conditional on η , $Y_1, \dots, Y_n$ is a random sample from the $N(\eta, \sigma^2)$ distribution, but that η has itself a $N(\mu, \sigma_\eta^2)$ distribution. Show that the unconditional distribution of $Y_1, \dots, Y_n$ is multivariate normal, with correlation $\rho = \sigma_\eta^2/(\sigma^2 + \sigma_\eta^2)$ between different variables.
(b) Show that

$$W = (\bar{Y} - \mu)/(\sigma^2/n)^{1/2} \stackrel{D}{=} \{1 + n\rho/(1 - \rho)\}^{1/2} T,$$

where $T \sim t_{n-1}$. Hence show that the probability that the usual confidence interval (3.17) contains μ is $1 - 2\Pr\{T \leq t_{n-1}(\alpha)(1 + n\sigma_\eta^2/\sigma^2)^{-1/2}\}$ and verify that when $\alpha = 0.025$, $n = 10$ and $\rho = 0.1$, this probability is 0.85, and that when $n = 100$ and $\rho = 0.01, 0.02$, it is 0.84, 0.74.

Figure 3.9 Samples from bivariate distributions with correlations $-0.7, 0, 0.7$; one sample has independent components. Which is which? Why?



What does this tell you about the assumptions underlying (3.17)?

- 5 If Z is standard normal, then $Y = \exp(\mu + \sigma Z)$ is said to have the log-normal distribution. Show that $E(Y^r) = \exp(r\mu)M_Z(r\sigma)$ and hence give expressions for the mean and variance of Y . Show that although all its moments are finite, Y does not have a moment-generating function.
- 6 (a) Let $Y = Z_1$ and $W = Z_2 - \lambda Z_1$, where Z_1, Z_2 are independent standard normal variables and λ is a real number. Show that the conditional density of Y given that $W < 0$ is $f(y; \lambda) = 2\phi(y)\Phi(\lambda y)$; Y is said to have a *skew-normal distribution*. Sketch $f(y; \lambda)$ for various values of λ . What happens when $\lambda = 0$?
 (b) Show that $Y^2 \sim \chi_1^2$.
 (c) Use Exercise 3.2.16 to show that Y has cumulant-generating function $t^2/2 + \log \Phi(\delta t)$, where $\delta = \lambda/(1 + \lambda^2)^{1/2}$, and hence find its mean and variance. Show that the standardized skewness of Y varies in the range $(-0.995, 0.955)$.
- 7 For $h(x)$ a non-negative function of real x with finite integral, let

$$C_h = \{(u, v) : 0 \leq u \leq h(v/u)^{1/2}\}.$$

(a) By considering the change of variables $(u, v) \rightarrow (w = u, x = v/u)$, show that C_h has finite area, and that if (U, V) is uniformly distributed on C_h , then $X = V/U$ has density $h(x)/\int h(y) dy$.

(b) If $h(x)$ and $x^2 h(x)$ are bounded and $a = \sqrt{\sup\{h(x) : -\infty < x < \infty\}}$,

$$b_+ = \sqrt{\sup\{x^2 h(x) : x \geq 0\}}, \quad b_- = -\sqrt{\sup\{x^2 h(x) : x \leq 0\}},$$

show that $C_h \subset [0, a] \times [b_-, b_+]$. Hence justify the following algorithm:

1 Repeat

- generate $U_1, U_2 \stackrel{\text{iid}}{\sim} U(0, 1)$;
 - let $U = aU_1, V = b_- + (b_+ - b_-)U_2$;
- until $(U, V) \in C_h$.

2 Return $X = V/U$.

(c) If $h(x) = (1 + x^2)^{-1}$ on $-\infty < x < \infty$, show that this algorithm gives the method for generating Cauchy variables described in Example 3.21.

(d) If $h(x) = e^{-x}$ on $0 < x < \infty$, show that $a = 1, b_- = 0$, and $b_+ = 2/e$, and give the algorithm.

(e) If $h(x) = e^{-x^2/2}$ on $-\infty < x < \infty$, find the values of a, b_- and b_+ , and show that X is accepted if and only if $V^2 \leq -4U^2 \log U$. Hence give the algorithm.

- 8 Let R_1, R_2 be independent binomial random variables with probabilities π_1, π_2 and denominators m_1, m_2 , and let $P_i = R_i/m_i$. It is desired to test if $\pi_1 = \pi_2$. Let $\hat{\pi} = (m_1 P_1 + m_2 P_2)/(m_1 + m_2)$. Show that when $\pi_1 = \pi_2$, the statistic

$$Z = \frac{P_1 - P_2}{\sqrt{\hat{\pi}(1 - \hat{\pi})(1/m_1 + 1/m_2)}} \xrightarrow{D} N(0, 1)$$

when $m_1, m_2 \rightarrow \infty$ in such a way that $m_1/m_2 \rightarrow \xi$ for $0 < \xi < 1$.

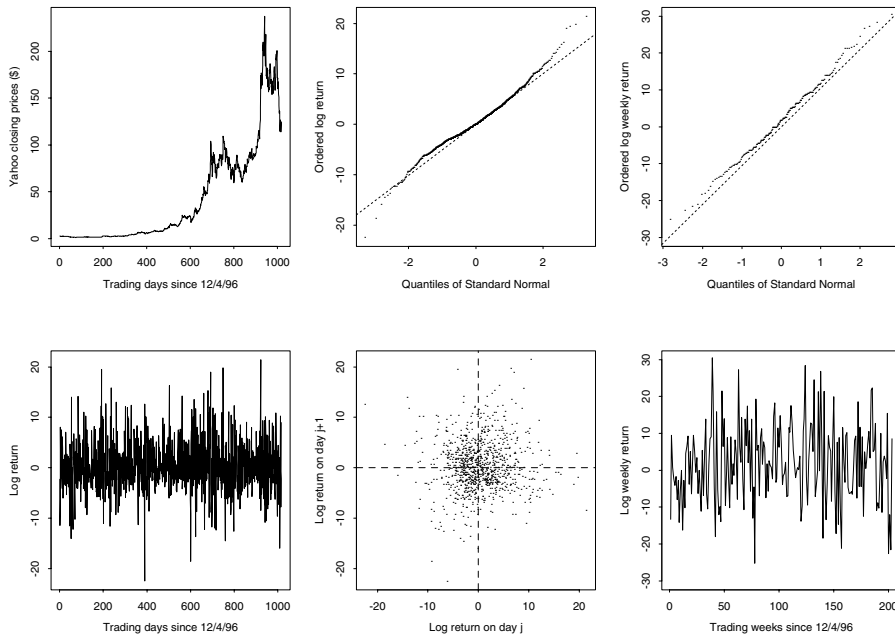


Figure 3.10 Analysis of Yahoo.com share values. Left: share price x_j from 12 April 1996 to 26 April 2000 (above); log daily returns $y_j = 100 \log(x_j/x_{j-1})$ (below). Centre: normal probability plot of y_j (above) and plot of y_{j+1} against y_j (below). Right: normal probability plot of log weekly returns (above); log weekly returns (below).

Now consider a 2×2 table formed using two independent binomial variables and having entries R_i, S_i where $R_i + S_i = m_i$, $R_i/m_i = P_i$, for $i = 1, 2$. Show that if $\pi_1 = \pi_2$ and $m_1, m_2 \rightarrow \infty$, then

$$X^2 = (n_1 + n_2)(R_1 S_2 - R_2 S_1)^2 / \{n_1 n_2 (R_1 + R_2)(S_1 + S_2)\} \xrightarrow{D} \chi_1^2.$$

Two batches of trees were planted in a park: 250 were obtained from nursery *A* and 250 from nursery *B*. Subsequently 41 and 64 trees from the two groups die. Do trees from the two nurseries have the same survival probabilities? Are the assumptions you make reasonable?

- 9 If $\bar{Y}$ is the average of a random sample $Y_1, \dots, Y_n$ from density $\theta^{-1} \exp(-y/\theta)$, $y > 0$, $\theta > 0$, give the limiting distribution of $Z(\theta) = n^{1/2}(\bar{Y} - \theta)/\theta$ as $n \rightarrow \infty$. Hence obtain an approximate two-sided 95% confidence interval for θ . Show that for large n , $\log(\bar{Y}) \doteq \log \theta + n^{-1/2} Z$, find an approximate mean and variance for $\log \bar{Y}$, and hence give another approximate two-sided 95% confidence interval for θ . Which interval would you prefer in practice?
- 10 Independent pairs (X_j, Y_j) , $j = 1, \dots, m$ arise in such a way that X_j is normal with mean λ_j and Y_j is normal with mean $\lambda_j + \psi$, X_j and Y_j are independent, and each has variance σ^2 . Find the joint distribution of $Z_1, \dots, Z_m$, where $Z_j = Y_j - X_j$, and hence show that there is a $(1 - 2\alpha)$ confidence interval for ψ of form $A \pm m^{-1/2} Bc$, where A and B are random variables and c is a constant. Obtain a 0.95 confidence interval for the mean difference ψ given (x, y) pairs (27, 26), (34, 30), (31, 31), (30, 32), (29, 25), (38, 35), (39, 33), (42, 32). Is it plausible that $\psi \neq 0$?
- 11 The upper left panel of Figure 3.10 shows daily closing share prices x_j for Yahoo.com from 12 April 1996 to 26 April 2000. We define the log daily returns $y_j = 100 \log(x_j/x_{j-1})$; y_j is roughly the daily percentage change in price.
 - (a) The lower left panel shows the y_j . Does their distribution seem to change with time?
 - (b) The upper central panel shows a normal probability plot of the y_j . Do they seem normal to you? If not, describe how they differ from normal variates.

(c) The lower central panel shows a plot of y_{j+1} against y_j . Are successive daily log returns correlated? What would be the implication if they were?

(d) The $n = 1015$ values of y_j have average and variance $\bar{y} = 0.376$ and $s^2 = 25.35$. Is $E(y_j) > 0$?

(e) We can also define the log weekly returns, $w_j = y_{5(j-1)+1} + \cdots + y_{5j}$, whose normal probability plot is shown in the top right panel. Are they normal? They have average and variance 1.878 and 110.07. Is their mean positive?

(f) The data suggest the simple *geometric Brownian motion* model that the stock value at the end of week k is $S_k = s_0 \exp\left(k\mu + \sigma \sum_{j=1}^k Z_j\right)$, where the Z_j are a standard normal random sample and s_0 is the initial stock value. If I bought \$100 worth of stock when it was launched and its value on 26 April 2000 was \$4527, give its median predicted value and a 95% prediction interval for its value 400 weeks after launch. Do you find this credible? Under the normal model, how long must I wait before the probability is 0.5 that I am a millionaire?

Remember: past performance is no guide to the future!

12 (a) Check the expressions for z_j for adaptive rejection sampling.

(b) Show that $G_+(y) = \int_{-\infty}^y g_+(x) dx$ satisfies

$$G_+(z_i) = \frac{\sum_{j=0}^i \frac{1}{h'(y_{j+1})} \{\exp h_+(z_{j+1}) - \exp h_+(z_j)\}}{\sum_{j=0}^{k-1} \frac{1}{h'(z_{j+1})} \{\exp h_+(z_{j+1}) - \exp h_+(z_j)\}};$$

let c_k denote the denominator of this expression. Show that a value X from g_+ is generated by taking $U \sim U(0, 1)$, finding the largest z_i such that $G_+(z_i) < U$ and setting

$$X = z_i + \frac{1}{h'(y_{i+1})} \log \left[1 + \frac{h'(y_{i+1})c_k \{U - G_+(z_i)\}}{\exp h_+(z_i)} \right].$$

(c) Let $h_-(y)$ be defined by taking the chords between the points $(y_j, h(y_j))$, for $j = 1, \dots, k$, and let it be $-\infty$ outside $[y_1, y_k]$. Explain how to use $h_-(y)$ to speed up sampling from f when h is complicated, by performing a pretest based on $\exp\{h_-(X) - h_+(X)\}$. (Gilks and Wild, 1992; Wild and Gilks, 1993)